

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

CHAPTER 2

Basic Topology

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

How to read this book

Everything set in the main column below is Rudin, verbatim. You can read it straight through and never look at an annotation. The annotation layer sits in two places, both outside his text.

Margin notes. A small superscript number in Rudin's text — like this¹ — keys to a numbered note in the outer margin. Since the book prints in black and white, the *tier* of a note is carried by how its number is set:

¹ *The marker is the only mark added to his page, and it is the same for every tier so his line spacing is undisturbed. The number out here tells you what kind of note it is.*

- 7** **Plain number — unpacking.** What the sentence is actually saying, where a formula came from, an omitted algebraic step.
- 7** **Outlined number — going deeper.** Context beyond the course: what breaks in general, which later theorem this sets up. Skippable.
- 7** **Reversed number — warning.** A specific belief students form here that is false, or a place where Rudin's wording will mislead you.

Interleaved boxes. When a note runs longer than a margin will hold, it becomes a box between Rudin's paragraphs, framed by the same three tiers: a **solid** left rule for unpacking, a **dashed** left rule for going deeper, and rules **top and bottom** with a black tab for a warning.

Every figure is an addition. Rudin's Chapter 1 contains no diagrams at all. Any picture you see here is annotation, drawn to expose geometry that his algebra states without illustrating; each carries a grey italic caption.

Where the text is corrected. The main column follows a retyped edition whose own foreword concedes that fresh typos were probably introduced. Where one is evident, the reading is restored from Rudin's third edition and a warning-tier note records the change — silent emendation would make “verbatim” meaningless.

Structural commentary. Two further boxes carry no mathematics at all. They explain the book rather than its contents — why a chapter exists, why Rudin ordered it as he did, why a section that looks like a detour is not one.

HEAVY FRAME, BLACK TITLE BAR

Opens and closes every chapter: what it is for, how it is organized and why in that order, which parts are load-bearing. Where Rudin has explained his own choices — usually in the Preface — these boxes quote him rather than guess.

THIN FRAME, INLINE HEADING | Opens a section: why this material sits here, and what to hold on to.

Books cited in the notes. Where an annotation says another treatment does something differently, it means one of these. They are named because seeing the same theorem set up two ways is often what makes Rudin’s choice legible.

<i>Tao, Analysis I–II</i>	Terence Tao, 3rd ed. Builds $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from the Peano axioms and constructs $\mathbb{R}$ by Cauchy sequences rather than cuts. Volume II covers metric spaces and uniform convergence — Rudin’s Chapters 2, 4 and 7.
<i>Yu Pin, Tsinghua notes</i>	数学分析讲义, Tsinghua / Yau Mathematical Sciences Center. Axiomatizes $\mathbb{R}$ by field, order, archimedean and <i>nested interval</i> axioms, then derives the supremum principle.
<i>ECNU text</i>	数学分析 (Mathematical Analysis), 5th ed., 2018, East China Normal University. Standard Chinese two-volume text. Its §7.1 collects six equivalent formulations of completeness and proves them interderivable.
<i>USTC notes</i>	中国科学技术大学少年班数学分析讲义 (程艺 et al., 2005). Teaches the main thread of calculus first and defers real-number theory to a later volume; lists Rudin among its three principal references.

Chapter 2

BASIC TOPOLOGY

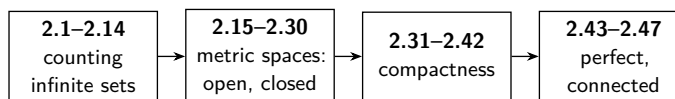
WHY THIS CHAPTER EXISTS

Chapter 1 secured a number system with no gaps. That is a statement about $\mathbb{R}$ alone, and almost nothing in analysis is about $\mathbb{R}$ alone — limits, continuity and integration all concern how points sit *near* one another. This chapter builds the language for nearness, and it builds it once, for a general metric space, so that $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^k$ and later spaces of functions are all covered by the same theorems.

The chapter's centre of gravity is **compactness** (2.31–2.42). It is the property that makes “infinitely many” behave like “finitely many,” and it is what will later force a continuous function on $[a, b]$ to be bounded (4.15), to attain its bounds (4.16), and to be uniformly continuous (4.19). If you take one thing from Chapter 2, take Theorem 2.41.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Five sections, and the first is not topology at all.



The first section is not topology. Countability is set theory, and Rudin puts it here because he needs it in two specific places: to know that $\mathbb{Q}$ is countable (used constantly), and to know that $\mathbb{R}$ is not (2.43, and the Cantor set at 2.44). You could read §§2.15–2.47 first and refer back.

The definitions are chosen for what they enable, not for what they describe. “Open,” “closed,” “limit point” will look like arbitrary vocabulary on first reading. They are not: each is exactly the hypothesis some later theorem needs. Watch where each is used rather than trying to intuit it in advance.

Compactness is defined by open covers (2.32), which is the strangest definition in the book and the one students resist hardest. Rudin’s reason appears only at 2.41, where he proves that for subsets of $\mathbb{R}^k$ it coincides with two more familiar

conditions. Until then, treat it as a formal property and watch what it does.

Everything is done for a general metric space where it can be, and for $\mathbb{R}^k$ where it cannot. That division is deliberate — 2.41 and 2.42 are *false* in general metric spaces, and knowing which theorems need $\mathbb{R}^k$ is worth more than knowing the theorems.

FINITE, COUNTABLE, AND UNCOUNTABLE SETS

WHAT THIS SECTION IS FOR | Two facts are needed later and everything here exists to produce them: $\mathbb{Q}$ is countable (Corollary to 2.13) and the set of 0–1 sequences is not (2.14). The first makes $\mathbb{Q}$ small enough to enumerate — spent in Exercises 2.18 and 2.29, in Chapter 3, and wherever a countable dense set is wanted. The second is what makes $\mathbb{R}$ genuinely bigger than $\mathbb{Q}$, and it reappears as 2.43.

The section also fixes notation for functions (2.1–2.2) that the whole book uses. Read 2.2 carefully even if the rest is familiar: Rudin’s f^{-1} is defined for *sets* and does not presuppose that f has an inverse.

We begin this section with a definition of the function concept.

2.1 Definition. Consider two sets A and B , whose elements may be any objects whatsoever, and suppose that with each element x of A there is associated, in some manner, an element of B , which we denote by $f(x)$. Then f is said to be a *function* from A to B (or a *mapping* of A into B). The set A is called the *domain* of f (we also say f is defined on A) and the elements $f(x)$ are called the *values* of f . The set of all values of f is called the *range* of f .²

2.2 Definition. Let A and B be two sets and let f be a mapping of A into B . If $E \subset A$, $f(E)$ is defined to be the set of all elements $f(x)$, for $x \in E$. We call $f(E)$ the *image* of E under f . In this notation, $f(A)$ is the range of f . It is clear that $f(A) \subset B$. If $f(A) = B$, we say that f maps A *onto* B . (Note that, according to this usage, *onto* is more specific than *into*.)

If $E \subset B$, $f^{-1}(E)$ denotes the set of all $x \in A$ such that $f(x) \in E$. We call $f^{-1}(E)$ the *inverse image* of E under f .³ If $y \in B$, $f^{-1}(y)$ is the set of all $x \in A$ such that $f(x) = y$. If, for each $y \in B$, $f^{-1}(y)$ consists of at most one element of A , then f is said to be a *1-1* (one-to-one) mapping of A into B . This may also be expressed as follows: f is a 1-1 mapping of A into B provided that $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$, $x_1 \in A$, $x_2 \in A$.

² “Associated, in some manner” is not a definition, and Rudin knows it — he is declining to build set theory, exactly as he declined to build $\mathbb{Z}$ in Chapter 1. One repair is to define a function as a set of ordered pairs in which each x appears exactly once; another, Tao’s (his 3.3.1), keeps functions primitive and defines them by any property $P(x, y)$ passing the “vertical line test,” with the ordered-pair encoding relegated to his Exercise 3.5.10. Nothing in this book depends on the choice.

³ f^{-1} here is an operation on subsets, and it is always defined — for every f , invertible or not. $f^{-1}(E)$ is simply “everything that lands in E ,” possibly empty. Do not read it as an inverse function. The clash of notation is universal and permanent, and the only defence is to check whether the argument is a set or a point.

(The notation $x_1 \neq x_2$ means that x_1 and x_2 are distinct elements; otherwise we write $x_1 = x_2$.)

Why inverse images, and not images, are the useful notion

Nothing yet suggests it, but f^{-1} is the better-behaved of the two operations, and Chapter 4 will turn on the fact. Inverse images respect every set operation:

$$\begin{aligned} f^{-1}(E \cup F) &= f^{-1}(E) \cup f^{-1}(F), \\ f^{-1}(E \cap F) &= f^{-1}(E) \cap f^{-1}(F), \\ f^{-1}(E^c) &= (f^{-1}(E))^c. \end{aligned}$$

Images respect only the first. $f(E \cap F) \subset f(E) \cap f(F)$ can be strict — take f constant and E, F disjoint — and images do not interact with complements at all.

That asymmetry is why Theorem 4.8 characterises continuity by “ f^{-1} of every open set is open” rather than by anything about images — no image-based version could work, since a constant map is continuous yet its images are single points, never open. Even the closure relation survives only one-sidedly for images: Exercise 4.2 proves $f(\overline{E}) \subset \overline{f(E)}$ and asks for an example where the inclusion is proper.

2.3 Definition. If there exists a 1-1 mapping of A onto B , we say that A and B can be put in *1-1 correspondence*, or that A and B have the same *cardinal number*, or, briefly, that A and B are *equivalent*, and we write $A \sim B$. This relation clearly has the following properties:

It is reflexive: $A \sim A$.

It is symmetric: If $A \sim B$, then $B \sim A$.

It is transitive: If $A \sim B$ and $B \sim C$, then $A \sim C$.

Any relation with these three properties is called an *equivalence relation*.

2.4 Definition. For any positive integer n , let $\mathbb{N}_n$ be the set whose elements are the integers $1, 2, \dots, n$; let $\mathbb{N}$ be the set consisting of all positive integers. For any set A , we say:

- (a) A is *finite* if $A \sim \mathbb{N}_n$ for some n (the empty set is also considered to be finite).
- (b) A is *infinite* if A is not finite.
- (c) A is *countable* if $A \sim \mathbb{N}$.
- (d) A is *uncountable* if A is neither finite nor countable.
- (e) A is *at most countable* if A is finite or countable.

Countable sets are sometimes called *enumerable*, or *denumerable*.⁴

For two finite sets A and B , we evidently have $A \sim B$ if and only if A and B contain the same number of elements. For infinite sets, however,

4 Rudin’s “countable” means countably infinite – it excludes finite sets. Many other books use “countable” for what he calls “at most countable,” and then say “countably infinite” for his sense.

When you read a theorem elsewhere, check which convention is in force; several of Rudin’s statements become false under the other one.

the idea of “having the same number of elements” becomes quite vague, whereas the notion of 1-1 correspondence retains its clarity.

2.5 Example. Let $\mathbb{Z}$ be the set of all integers. Then $\mathbb{Z}$ is countable. For, consider the following arrangement of the sets $\mathbb{Z}$ and $\mathbb{N}$:

$$\begin{array}{cccccccc} \mathbb{Z}: & 0, & 1, & -1, & 2, & -2, & 3, & -3, & \dots \\ \mathbb{N}: & 1, & 2, & 3, & 4, & 5, & 6, & 7, & \dots \end{array}$$

We can, in this example, even give an explicit formula for a function f from $\mathbb{N}$ to $\mathbb{Z}$ which sets up a 1-1 correspondence:

$$f(n) = \begin{cases} \frac{n}{2} & (n \text{ even}), \\ -\frac{n-1}{2} & (n \text{ odd}). \end{cases}$$

$$\begin{array}{ccccccccccc} \mathbb{Z} & 0 & & 1 & & -1 & & 2 & & -2 & & 3 & & -3 & & \dots \\ & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \dots \\ \mathbb{N} & 1 & & 2 & & 3 & & 4 & & 5 & & 6 & & 7 & & \dots \end{array}$$

Counting the integers by alternating sides of 0. Every integer is reached exactly once, which is all “countable” asks.

2.6 Remark. A finite set cannot be equivalent to one of its proper subsets. That this is, however, possible for infinite sets, is shown by Example 2.5, in which $\mathbb{N}$ is a proper subset of $\mathbb{Z}$.

In fact, we could replace Definition 2.4(b) by the statement: A is infinite if A is equivalent to one of its proper subsets.⁵

2.7 Definition. By a *sequence*, we mean a function f defined on the set $\mathbb{N}$ of all positive integers. If $f(n) = x_n$, for $n \in \mathbb{N}$, it is customary to denote the sequence f by the symbol $(x_n)_{n=1}^{\infty}$, or more briefly by (x_n) , or sometimes by $x_1, x_2, x_3, \dots$. The values of f , that is, the elements x_n , are called the *terms* of the sequence. If A is a set and if $x_n \in A$ for all $n \in \mathbb{N}$, then (x_n) is said to be a *sequence in A* , or a *sequence of elements of A* .

Note that the terms $x_1, x_2, x_3, \dots$ of a sequence need not be distinct.⁶

Since every countable set is the range of a 1-1 function defined on $\mathbb{N}$, we may regard every countable set as the range of a sequence of distinct terms. Speaking more loosely, we may say that the elements of any countable set can be “arranged in a sequence.”

Sometimes it is convenient to replace $\mathbb{N}$ in this definition by the set of all nonnegative integers, i.e., to start with 0 rather than 1.

2.8 Theorem. *Every infinite subset of a countable set A is countable.*

Proof. Suppose $E \subset A$, and E is infinite. Arrange the elements x of A in a sequence (x_n) of distinct elements. Construct a sequence (n_k) as follows:

Let n_1 be the smallest positive integer such that $x_{n_1} \in E$. Having chosen $n_1, \dots, n_{k-1}$ ($k = 2, 3, 4, \dots$), let n_k be the smallest integer greater than n_{k-1} such that $x_{n_k} \in E$.⁷

⁵ This is Dedekind’s definition of infinity, and taking it as primary is attractive because it needs no reference to $\mathbb{N}$ at all. The two definitions agree, but proving they agree requires a form of the axiom of choice — which is why Rudin offers the Dedekind version as an aside rather than adopting it.

⁶ A sequence is not its range. $(1, 1, 1, \dots)$ is an infinite sequence whose range is the one-element set $\{1\}$. Rudin flags this because the next paragraph slides between the two, and because Chapter 3 will repeatedly distinguish “the sequence converges” from statements about the set of its values.

⁷ “Smallest” is well-ordering again, the principle used silently in 1.1 and in the proof of 1.20(b). Choosing the least admissible index at each stage is what makes the construction canonical — no arbitrary choices are made, so no appeal to the axiom of choice is needed here. Contrast the proof of 2.12.

Putting $f(k) = x_{n_k}$ ($k = 1, 2, 3, \dots$), we obtain a 1-1 correspondence between E and $\mathbb{N}$. $\square$

The theorem shows that, roughly speaking, countable sets represent the “smallest” infinite: No uncountable set can be a subset of a countable set.

WHAT 2.8 RULES OUT

Read as a statement about size, 2.8 says there is nothing strictly between finite and countable. An infinite subset of a countable set cannot be a “smaller infinity” — it is countable, full stop.

That has an immediate consequence used everywhere: to prove some set S is countable it is enough to exhibit a 1-1 map from S into $\mathbb{N}$, or into any countable set, together with the knowledge that S is infinite. You never have to construct a bijection. Rudin uses exactly this shortcut at the end of 2.12, where he obtains $S \sim T$ for some $T \subset \mathbb{N}$ and then invokes 2.8.

2.9 Definition. Let A and Ω be sets, and suppose that with each element α of A there is associated a subset of Ω which we denote by E_α .

The set whose elements are the sets E_α will be denoted by $\{E_\alpha\}$. Instead of speaking of sets of sets, we shall sometimes speak of a *collection* of sets, or a *family* of sets.

The *union* of the sets E_α is defined to be the set S such that $x \in S$ if and only if $x \in E_\alpha$ for at least one $\alpha \in A$. We use the notation

$$S = \bigcup_{\alpha \in A} E_\alpha. \quad (2.1)$$

If A consists of the integers $1, 2, \dots, n$, one usually writes

$$S = \bigcup_{m=1}^n E_m \quad (2.2)$$

or

$$S = E_1 \cup E_2 \cup \dots \cup E_n. \quad (2.3)$$

If A is the set of all positive integers, the usual notation is

$$S = \bigcup_{m=1}^{\infty} E_m. \quad (2.4)$$

8 Worth pausing on. The ∞ in $\bigcup_{m=1}^{\infty}$ is not a limit and nothing is approached — it abbreviates “over all $m \in \mathbb{N}$,” and the union is defined outright by the membership condition, not as the limit of finite unions. Rudin flags the clash because Chapter 3 will use the same symbol for genuine limits.

The symbol ∞ in (2.4) merely indicates that the union of a countable collection of sets is taken, and should not be confused with the symbols $+\infty$, $-\infty$, introduced in Definition 1.23.⁸

The *intersection* of the sets E_α is defined to be the set P such that $x \in P$ if and only if $x \in E_\alpha$ for every $\alpha \in A$. We use the notation

$$P = \bigcap_{\alpha \in A} E_\alpha, \quad (2.5)$$

or

$$P = \bigcap_{m=1}^n E_m = E_1 \cap E_2 \cap \cdots \cap E_n, \quad (2.6)$$

or

$$P = \bigcap_{m=1}^{\infty} E_m, \quad (2.7)$$

as for unions. If $A \cap B$ is not empty, we say that A and B *intersect*; otherwise they are *disjoint*.

2.10 Examples.

(a) Suppose E_1 consists of 1, 2, 3 and E_2 consists of 2, 3, 4. Then $E_1 \cup E_2$ consists of 1, 2, 3, 4, whereas $E_1 \cap E_2$ consists of 2, 3.

(b) Let A be the set of real numbers x such that $0 < x \leq 1$. For every $x \in A$, let E_x be the set of real numbers y such that $0 < y < x$. Then

(i) $E_x \subset E_z$ if and only if $0 < x \leq z \leq 1$;

(ii) $\bigcup_{x \in A} E_x = E_1$;

(iii) $\bigcap_{x \in A} E_x$ is empty.

(i) and (ii) are clear. To prove (iii), we note that for every $y > 0$, $y \notin E_x$ if $x < y$. Hence $y \notin \bigcap_{x \in A} E_x$.⁹

2.11 Remarks. Many properties of unions and intersections are quite similar to those of sums and products; in fact, the words *sum* and *product* were sometimes used in this connection, and the symbols Σ and Π were written in place of $\bigcup$ and $\bigcap$.

The commutative and associative laws are trivial:

$$A \cup B = B \cup A; \quad A \cap B = B \cap A. \quad (2.8)$$

$$(A \cup B) \cup C = A \cup (B \cup C); \quad (A \cap B) \cap C = A \cap (B \cap C). \quad (2.9)$$

Thus the omission of parentheses in (2.3) and (2.6) is justified.

The distributive law also holds:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C). \quad (2.10)$$

To prove this, let the left and right members of (2.10) be denoted by E and F , respectively.

Suppose $x \in E$. Then $x \in A$ and $x \in B \cup C$, that is, $x \in B$ or $x \in C$ (possibly both). Hence $x \in A \cap B$ or $x \in A \cap C$, so that $x \in F$. Thus $E \subset F$.

Next, suppose $x \in F$. Then $x \in A \cap B$ or $x \in A \cap C$. That is, $x \in A$, and $x \in B \cup C$. Hence $x \in A \cap (B \cup C)$, so that $F \subset E$.

It follows that $E = F$.¹⁰

⁹ Example (b)(iii) is the one to remember: an intersection of nonempty nested sets can be empty. Each E_x contains points, and any finitely many of them have a common point, yet nothing survives all of them. Theorem 2.36 will show that compactness is exactly what forbids this, and 2.39 is the version for k -cells.

¹⁰ The template for every set identity in the book: prove two inclusions, each by chasing a single element. Rudin writes it out once here and never again — later identities are left as “easily verified.”

We list a few more relations which are easily verified:

$$A \subset A \cup B, \quad (2.11)$$

$$A \cap B \subset A. \quad (2.12)$$

If $\emptyset$ denotes the empty set, then

$$A \cup \emptyset = A, \quad A \cap \emptyset = \emptyset. \quad (2.13)$$

If $A \subset B$, then

$$A \cup B = B, \quad A \cap B = A. \quad (2.14)$$

2.12 Theorem. Let $\{E_n\}$, $n = 1, 2, 3, \dots$, be a countable collection of countable sets, and put

$$S = \bigcup_{n=1}^{\infty} E_n. \quad (2.15)$$

Then S is countable.

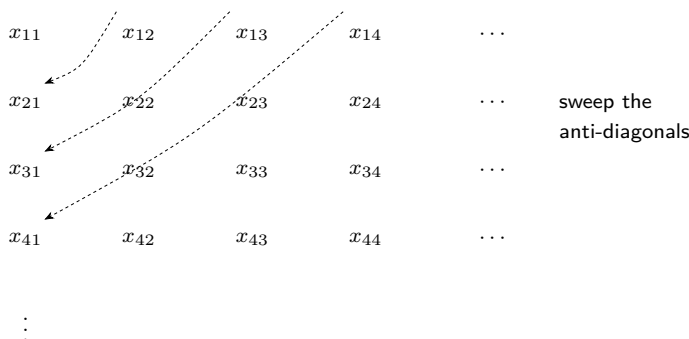
Proof. Let every set E_n be arranged in a sequence (x_{nk}) , $k = 1, 2, 3, \dots$, and consider the infinite array¹¹

$$\begin{array}{cccccc} x_{11} & x_{12} & x_{13} & x_{14} & \cdots \\ x_{21} & x_{22} & x_{23} & x_{24} & \cdots \\ x_{31} & x_{32} & x_{33} & x_{34} & \cdots \\ x_{41} & x_{42} & x_{43} & x_{44} & \cdots \\ \dots & \dots & \dots & \dots & \dots \end{array} \quad (2.16)$$

in which the elements of E_n form the n th row. The array contains all elements of S . As indicated by the arrows, these elements can be arranged in a sequence

$$x_{11}; x_{21}, x_{12}; x_{31}, x_{22}, x_{13}; x_{41}, x_{32}, x_{23}, x_{14}; \dots \quad (2.17)$$

If any two of the sets E_n have elements in common, these will appear more than once in (2.17). Hence there is a subset T of the set of all positive integers such that $S \sim T$, which shows that S is at most countable (Theorem 2.8). Since $E_1 \subset S$, and E_1 is infinite, S is infinite, and thus countable. $\square$



Array (2.16). Each anti-diagonal is finite, so sweeping them in turn reaches every entry after finitely many steps — which is exactly what an enumeration must do.

11 This opening clause is where the axiom of countable choice enters, and Rudin does not mention it. Each E_n can be arranged in a sequence, but there is no canonical way to do it, and we are choosing one arrangement for each of infinitely many n simultaneously. Without some choice principle the theorem is not provable — it is consistent with ZF that $\mathbb{R}$ is a countable union of countable sets. Compare 2.8, where “smallest index” made every choice canonical and no such principle was needed.

WHY THE DIAGONAL SWEEP, AND NOT THE OBVIOUS ORDERS

The natural first attempts both fail, and seeing why is the content of the proof.

Row by row never finishes row 1, so x_{21} is never reached. *Column by column* fails the same way. What the anti-diagonals have that rows do not is **finiteness**: the d th anti-diagonal holds exactly d entries, so every entry sits at a finite distance from the start.

That is the whole idea, and it is worth stating in the form used later: a countable union of *finite* sets is at most countable, and the array argument works by exhibiting S as such a union — the anti-diagonals.

Corollary. Suppose A is at most countable, and, for every $\alpha \in A$, B_α is at most countable. Put

$$T = \bigcup_{\alpha \in A} B_\alpha.$$

Then T is at most countable, for T is equivalent to a subset of (2.15).

2.13 Theorem. Let A be a countable set, and let B_n be the set of all n -tuples $(a_1, \dots, a_n)$, where $a_k \in A$ ($k = 1, \dots, n$), and the elements $a_1, \dots, a_n$ need not be distinct. Then B_n is countable.

Proof. That B_1 is countable is evident, since $B_1 = A$. Suppose B_{n-1} is countable ($n = 2, 3, 4, \dots$). The elements of B_n are of the form

$$(b, a) \quad (b \in B_{n-1}, a \in A). \quad (2.18)$$

For every fixed b , the set of pairs (b, a) is equivalent to A , and hence countable. Thus B_n is the union of a countable set of countable sets. By Theorem 2.12, B_n is countable.

The theorem follows by induction. $\square$

Corollary. The set of all rational numbers is countable.

Proof. We apply Theorem 2.13, with $n = 2$, noting that every rational r is of the form b/a , where a and b are integers. The set of pairs (a, b) , and therefore the set of fractions b/a , is countable. $\square$

The corollary is the reason for the section

$\mathbb{Q}$ countable is used more often than any other fact in this chapter. It gives $\mathbb{R}$ a countable dense subset (by 1.20(b)) — which is what Exercise 2.18 uses to build a perfect set avoiding every rational, what the structure theorem for open sets (Exercise 2.29) rests on, and what makes $\mathbb{R}^k$ separable (Exercise 2.22).

Note also the shape of the argument, which recurs: $\mathbb{Q}$ is not obviously countable, but it is the image of a set of *pairs*, and pairs are handled by 2.13. Whenever an object is specified by finitely many integers, this argument shows there are only countably many

of them — which is how Exercise 2.2 gets the algebraic numbers. The strict inequality $|\mathbb{Q}| < |\mathbb{R}|$ that follows from 2.14 is worth pausing on for what it says about 1.20(b): between any two reals there is a rational, yet the rationals are vanishingly rare. Density and abundance are unrelated.

In fact, even the set of all algebraic numbers is countable (see Exercise 2.2). That not all infinite sets are, however, countable, is shown by the next theorem.

2.14 Theorem. *Let A be the set of all sequences whose elements are the digits 0 and 1. This set A is uncountable.*

The elements of A are sequences like $1, 0, 0, 1, 0, 1, 1, 1, \dots$

Proof. Let E be a countable subset of A , and let E consist of the sequences $s_1, s_2, s_3, \dots$. We construct a sequence s as follows. If the n th digit in s_n is 1, we let the n th digit of s be 0, and vice versa. Then the sequence s differs from every member of E in at least one place; hence $s \notin E$. But clearly $s \in A$, so that E is a proper subset of A .

We have shown that every countable subset of A is a proper subset of A . It follows that A is uncountable (for otherwise A would be a proper subset of A , which is absurd).¹² $\square$

¹² The last step is where 2.6 is spent. If A were countable then A itself would be one of the countable subsets E , and the argument just showed every such E is a proper subset of A — so $A \subsetneq A$. Rudin's phrasing ("every countable subset is proper") avoids having to fix an enumeration of A in advance and then derive a contradiction, which is the more familiar but slightly clumsier route.

s_1	1	1	0	1	0	$\dots$	
s_2	0	1	1	0	1	$\dots$	
s_3	1	0	0	0	1	$\dots$	
s_4	1	1	1	0	0	$\dots$	
s_5	0	0	1	1	1	$\dots$	
s	0	0	1	1	0	$\dots$	flip the boxed diagonal

Cantor's diagonal process. The new sequence s disagrees with s_n in place n , so it is not in the list — whatever the list was.

The idea of the above proof was first used by Cantor, and is called *Cantor's diagonal process*; for, if the sequences $s_1, s_2, s_3, \dots$ are placed in an array like (2.16), it is the elements on the diagonal which are involved in the construction of the new sequence.

Readers who are familiar with the binary representation of the real numbers (base 2 instead of 10) will notice that Theorem 2.14 implies that the set of all real numbers is uncountable. We shall give a second proof of this fact in Theorem 2.43.

THE BINARY ARGUMENT NEEDS ONE REPAIR

The map from 0–1 sequences to reals in $[0, 1]$ is *not* a bijection: $0.0111\dots$ and $0.1000\dots$ name the same number, exactly as $0.999\dots = 1$ in base ten. So 2.14 does not immediately transfer.

The repair is cheap. Only the dyadic rationals have two expansions, and there are countably many of them, so the map fails to be injective on a countable set — not enough to make an uncountable set countable. But the gap is real, and it is the same delicacy that made Rudin warn “we shall never use decimals” at 1.22 and made Tao call the decimal construction of $\mathbb{R}$ the most complicated of the three.

Theorem 2.43 avoids the issue entirely, which is presumably why Rudin promises a second proof.

METRIC SPACES

WHY THIS SECTION IS HERE | Chapter 1 ended by noting that $\mathbb{R}^k$ has a distance function satisfying three properties (1.37(a),(b),(f)). This section takes those three properties as the *definition* of a space and develops everything that follows from them alone. The gain is reach: every theorem proved here holds for $\mathbb{R}$, for $\mathbb{C}$, for $\mathbb{R}^k$, for subsets of any of these, and later for spaces of functions (7.14, 11.35).

Definition 2.18 is the hinge of the chapter — ten pieces of vocabulary in one block, arriving with no motivation. Do not try to absorb them all. The two that carry the weight are **limit point** (b) and **open** (f); “closed” is defined from the first, and everything in 2.19–2.30 is bookkeeping about how the two interact. The rest of the list is there so Rudin need not interrupt himself later.

Read 2.29–2.30 with attention even though they look technical. They say that “open” is not a property a set has by itself but one it has *relative to an ambient space*, and that ambiguity causes more confusion in Chapter 4 than any other point in the book.

2.15 Definition. A set X , whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p, q)$, called the *distance* from p to q , such that

- (a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;
- (b) $d(p, q) = d(q, p)$;
- (c) $d(p, q) \leq d(p, r) + d(r, q)$ for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.¹³

2.16 Examples. The most important examples of metric spaces, from our standpoint, are the euclidean spaces $\mathbb{R}^k$, especially $\mathbb{R}^1$ (the real line) and $\mathbb{R}^2$ (the complex plane); the distance in $\mathbb{R}^k$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^k). \quad (2.19)$$

13 Compare 1.37: (a) here is 1.37(a),(b); (b) follows from 1.37(c) with $\alpha = -1$; (c) is 1.37(f). Rudin verified all three for $\mathbb{R}^k$ one chapter ago, which is why 2.16 can assert the example in a single line. Note what is absent: no addition, no scalar multiplication, no inner product. A metric space has distance and nothing else.

By Theorem 1.37, the conditions of Definition 2.15 are satisfied by (2.19).

It is important to observe that every subset Y of a metric space X is a metric space in its own right, with the same distance function. For it is clear that if conditions (a) to (c) of Definition 2.15 hold for $p, q, r \in X$, they also hold if we restrict p, q, r to lie in Y .¹⁴

Thus, every subset of a euclidean space is a metric space. Other examples are the spaces $\mathcal{C}(K)$ and $\mathcal{L}^2(\mu)$, which are discussed in Chaps. 7 and 11, respectively.

2.17 Definition. By the *segment* (a, b) we mean the set of all real numbers x such that $a < x < b$.

By the *interval* $[a, b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

Occasionally we shall also encounter “half-open intervals” $[a, b)$ and $(a, b]$; the first consists of all x such that $a \leq x < b$, the second of all x such that $a < x \leq b$.

If $a_i < b_i$ for $i = 1, \dots, k$, the set of all points $\mathbf{x} = (x_1, \dots, x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ ($1 \leq i \leq k$) is called a *k-cell*. Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.¹⁵

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$ (or $|\mathbf{y} - \mathbf{x}| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

For example, balls are convex. For if $|\mathbf{y} - \mathbf{x}| < r$, $|\mathbf{z} - \mathbf{x}| < r$, and $0 < \lambda < 1$, we have

$$\begin{aligned} |\lambda \mathbf{y} + (1 - \lambda) \mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\ &\leq \lambda |\mathbf{y} - \mathbf{x}| + (1 - \lambda) |\mathbf{z} - \mathbf{x}| < \lambda r + (1 - \lambda) r \\ &= r. \end{aligned}$$

The same proof applies to closed balls. It is also easy to see that *k*-cells are convex.

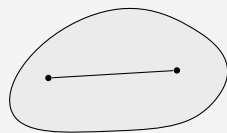
Convexity is not a metric notion

Notice that convexity is defined only for $E \subset \mathbb{R}^k$, never for a general metric space — and it could not be, because $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ requires addition and scalar multiplication, which 2.15 does not provide. Convexity belongs to the vector space structure, not the metric. Rudin introduces it here and then barely uses it in this chapter — Exercise 2.21(c) below proves convex sets are connected, and the next real use is Theorem 9.19, where the mean value inequality needs the segment joining two points to stay inside the domain. (Convex *functions*, a different notion, appear in Exercise 4.23.) The

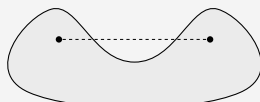
¹⁴ This one-sentence observation is what makes 2.29–2.30 necessary. Because every subset is itself a metric space, the words “open” and “closed” become ambiguous the moment more than one ambient space is in play — and from Chapter 4 onward there always is.

¹⁵ Rudin’s “segment” means open interval and his “interval” means closed. Almost no other book uses these words this way, and he uses them constantly. A *k*-cell is always closed — the inequalities are $\leq$ — which is why 2.40 can prove they are compact.

line “balls are convex” is worth keeping, though: it depends on the triangle inequality alone and is the reason pictures of neighbourhoods may be drawn as round blobs without loss.



convex: every segment stays in



not convex: this one leaves

The definition quantifies over every pair of points and every λ : one escaping chord is enough to fail it.

2.18 Definition. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X .

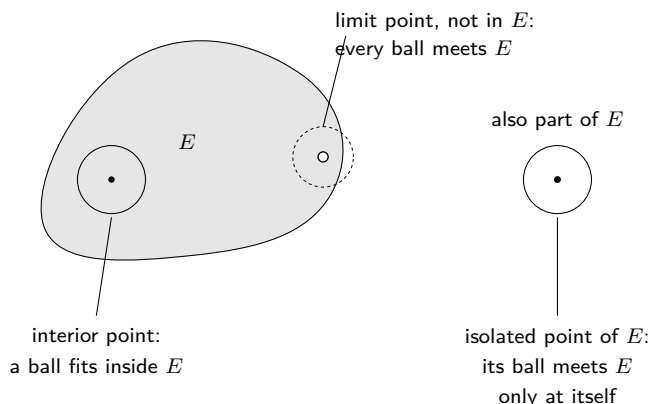
- (a) A *neighborhood* of p is a set $N_r(p)$ consisting of all q such that $d(p, q) < r$ for some $r > 0$. The number r is called the *radius* of $N_r(p)$.
- (b) A point p is a *limit point* of the set E if every neighborhood of p contains a point $q \neq p$ such that $q \in E$.¹⁶
- (c) If $p \in E$ and p is not a limit point of E , then p is called an *isolated point* of E .
- (d) E is *closed* if every limit point of E is a point of E .
- (e) A point p is an *interior point* of E if there is a neighborhood N of p such that $N \subset E$.
- (f) E is *open* if every point of E is an interior point of E .
- (g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.
- (h) E is *perfect* if E is closed and if every point of E is a limit point of E .
- (i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p, q) < M$ for all $p \in E$.
- (j) E is *dense* in X if every point of X is a limit point of E , or a point of E (or both).

Let us note that in $\mathbb{R}^1$ neighborhoods are segments, whereas in $\mathbb{R}^2$ neighborhoods are interiors of circles.

THE TEN DEFINITIONS IN ONE PICTURE

Take E to be a shaded blob in the plane together with one stray point off to the side. Every term in 2.18 can be read off it.

16 The clause $q \neq p$ is the whole definition. Without it every point of E would trivially be a limit point of E , and the notion would be useless. A limit point is a point that E crowds from outside itself — and it need not belong to E : see 2.21(e), where 0 is a limit point of $\{1/n\}$ and lies nowhere in the set.



A neighbourhood is a ball. An interior point has one inside E ; a limit point has every one of its balls meeting E elsewhere; an isolated point of E has a ball meeting E only at itself.

Three consequences are worth fixing now, because Rudin uses them without comment.

Open and closed are not opposites. They are defined by unrelated conditions — (f) is about points *of* E , (d) is about limit points, which may lie outside. A set can be both (the whole space X , and $\emptyset$), or neither (the half-open $[0, 1)$ in $\mathbb{R}$). Theorem 2.23 supplies the only link there is: E open $\iff E^c$ closed.

A point of E is either isolated or a limit point of E , by (c), and never both. A perfect set (h) is one with no isolated points at all, closed.

“Dense” means the closure is everything. Once 2.26 defines $\overline{E} = E \cup E'$, clause (j) reads $\overline{E} = X$. Theorem 1.20(b) says exactly that $\mathbb{Q}$ is dense in $\mathbb{R}$.

2.19 Theorem. *Every neighborhood is an open set.*

Proof. Consider a neighborhood $E = N_r(p)$, and let q be any point of E . Then there is a positive real number h such that

$$d(p, q) = r - h.$$

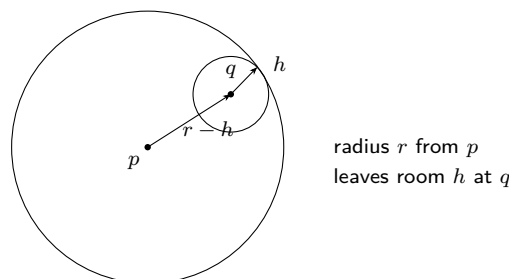
For all points s such that $d(q, s) < h$, we have then

$$d(p, s) \leq d(p, q) + d(q, s) < r - h + h = r,$$

so that $s \in E$. Thus q is an interior point of E . $\square$

A BALL INSIDE A BALL

The statement sounds like a tautology and is not: “neighborhood” is defined by a distance inequality, “open” by the existence of small balls around every point. The theorem says the first implies the second, and the triangle inequality is what does it.



Whatever is left over after travelling $d(p, q)$ is exactly the radius that fits at q .

The number $h = r - d(p, q)$ is positive precisely because q is *inside* the neighbourhood. That is the only thing being used, and it is why closed balls in $\mathbb{R}^k$ are not open: a point on the boundary sphere has $h = 0$ and no room at all.

2.20 Theorem. *If p is a limit point of a set E , then every neighborhood of p contains infinitely many points of E .*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E . Let $q_1, \dots, q_n$ be those points of $N \cap E$ which are distinct from p , and put

$$r = \min_{1 \leq m \leq n} d(p, q_m).$$

[We use this notation to denote the smallest of the numbers $d(p, q_1), \dots, d(p, q_n)$.] The minimum of a finite set of positive numbers is clearly positive, so that $r > 0$.¹⁷

The neighborhood $N_r(p)$ contains no point q of E such that $q \neq p$, so that p is not a limit point of E . This contradiction establishes the theorem. $\square$

Corollary. *A finite point set has no limit points.*

2.21 Examples. Let us consider the following subsets of $\mathbb{R}^2$:

- (a) The set of all complex z such that $|z| < 1$.
- (b) The set of all complex z such that $|z| \leq 1$.
- (c) A nonempty finite set.
- (d) The set of all integers.
- (e) The set consisting of the numbers $1/n$ ($n = 1, 2, 3, \dots$). Let us note that this set E has a limit point (namely, $z = 0$) but that no point of E is a limit point of E ; we wish to stress the difference between having a limit point and containing one.
- (f) The set of all complex numbers (that is, $\mathbb{R}^2$).
- (g) The segment (a, b) .

¹⁷ Here is where finiteness is spent, and it is the entire proof. An infinite set of positive numbers can have infimum 0 — take $\{1/n\}$ — so no such r could be extracted. The same finite-minimum move appears in 2.24(c), and both are the reason those theorems say “finite.”

Let us note that (d), (e), (g) can be regarded also as subsets of $\mathbb{R}^1$.

Some properties of these sets are tabulated below:

	Closed	Open	Perfect	Bounded
(a)	No	Yes	No	Yes
(b)	Yes	No	Yes	Yes
(c)	Yes	No	No	Yes
(d)	Yes	No	No	No
(e)	No	No	No	Yes
(f)	Yes	Yes	Yes	No
(g)	No		No	Yes

In (g), we left the second entry blank. The reason is that the segment (a, b) is not open if we regard it as a subset of $\mathbb{R}^2$, but it is an open subset of $\mathbb{R}^1$.¹⁸

18 The blank entry is the most instructive cell in the table. “Open” is not a property of a set; it is a property of a set in an ambient space. As a subset of the line, (a, b) is open; the same points sitting in the plane form a segment with empty interior, since no disc fits inside it. This is what 2.29–2.30 are about, and forgetting it is the standard error in Chapter 4.

READING THE TABLE

Row (f) is worth staring at: $\mathbb{R}^2$ is closed *and* open in itself. So is $\emptyset$. That is not a paradox — it is what the definitions say. In fact connectedness (2.45) amounts to exactly this: a space is connected precisely when $\emptyset$ and the whole space are its only subsets that are both open and closed, as a short argument from Definition 2.45 shows.

Row (e) is the one Rudin flags. $E = \{1/n\}$ has the limit point 0 but does not *contain* it, so E is not closed. No point of E is a limit point of E , so every point of E is isolated. A set can therefore be entirely isolated points and still have a limit point elsewhere.

Rows (b) and (f) are the perfect ones. Perfect means closed with no isolated points, and (c),(d),(e) fail it for having isolated points, while (a) and (g) fail it for not being closed.

2.22 Theorem. Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α . Then

$$\left(\bigcup_{\alpha} E_{\alpha}\right)^c = \bigcap_{\alpha} (E_{\alpha}^c). \quad (2.20)$$

Proof. Let A and B be the left and right members of (2.20). If $x \in A$, then $x \notin \bigcup_{\alpha} E_{\alpha}$, hence $x \notin E_{\alpha}$ for any α , hence $x \in E_{\alpha}^c$ for every α , so that $x \in \bigcap_{\alpha} E_{\alpha}^c$. Thus $A \subset B$.

Conversely, if $x \in B$, then $x \in E_{\alpha}^c$ for every α , hence $x \notin E_{\alpha}$ for any α , hence $x \notin \bigcup_{\alpha} E_{\alpha}$, so that $x \in \left(\bigcup_{\alpha} E_{\alpha}\right)^c$. Thus $B \subset A$.

It follows that $A = B$. $\square$

2.23 Theorem. A set E is open if and only if its complement is closed.

Proof. First, suppose E^c is closed. Choose $x \in E$. Then $x \notin E^c$, and x is not a limit point of E^c . Hence there exists a neighborhood N of x

such that $E^c \cap N$ is empty, that is, $N \subset E$. Thus x is an interior point of E , and E is open.

Next, suppose E is open. Let x be a limit point of E^c . Then every neighborhood of x contains a point of E^c , so that x is not an interior point of E . Since E is open, this means that $x \in E^c$. It follows that E^c is closed. $\square$

Corollary. *A set F is closed if and only if its complement is open.*

2.22 and 2.23 are the engine of the section

Every remaining proof about closed sets is obtained by complementing a proof about open sets. 2.22 (De Morgan) turns unions into intersections; 2.23 turns open into closed. Together they let Rudin prove half of 2.24 and get the other half free, which is exactly what he does.

The pattern is worth naming because it recurs throughout the book: prove the statement for open sets, where the definition is local and easy to verify at a point, then complement. Closedness is a statement about limit points, which are harder to handle directly — so one almost never argues with it head on.

2.24 Theorem.

- (a) *For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.*
- (b) *For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.*
- (c) *For any finite collection $G_1, \dots, G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.*
- (d) *For any finite collection $F_1, \dots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.*

Proof. Put $G = \bigcup_\alpha G_\alpha$. If $x \in G$ then $x \in G_\alpha$ for some α . Since x is an interior point of G_α , x is also an interior point of G , and G is open. This proves (a).

By Theorem 2.22,

$$\left(\bigcap_\alpha F_\alpha\right)^c = \bigcup_\alpha (F_\alpha^c), \quad (2.21)$$

and F_α^c is open, by Theorem 2.23. Hence (a) implies that (2.21) is open, so that $\bigcap_\alpha F_\alpha$ is closed.

Next, put $H = \bigcap_{i=1}^n G_i$. For any $x \in H$, there exist neighborhoods N_i of x , with radii r_i , such that $N_i \subset G_i$ ($i = 1, \dots, n$). Put

$$r = \min\{r_1, \dots, r_n\}$$

and let N be the neighborhood of x of radius r . Then $N \subset G_i$ for $i = 1, \dots, n$, so that $N \subset H$, and H is open.¹⁹

By taking complements, (d) follows from (c):

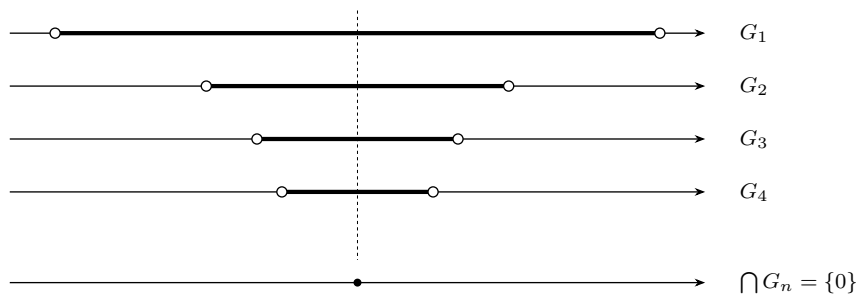
$$\left(\bigcup_{i=1}^n F_i\right)^c = \bigcap_{i=1}^n (F_i^c).$$

19 The finite minimum again, exactly as in 2.20. With infinitely many G_i the radii could shrink to zero and leave no room, which is precisely what happens in Example 2.25.

□

2.25 Example. In parts (c) and (d) of the preceding theorem, the finiteness of the collections is essential. For let G_n be the segment $(-1/n, 1/n)$ ($n = 1, 2, 3, \dots$). Then G_n is an open subset of $\mathbb{R}^1$. Put $G = \bigcap_{n=1}^{\infty} G_n$. Then G consists of a single point (namely, $x = 0$) and is therefore not an open subset of $\mathbb{R}^1$.

Thus the intersection of an infinite collection of open sets need not be open. Similarly, the union of an infinite collection of closed sets need not be closed.



Each G_n is open, yet the radii shrink to nothing and the intersection is a single point — not open.

2.26 Definition. If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X , then the *closure* of E is the set $\overline{E} = E \cup E'$.

2.27 Theorem. If X is a metric space and $E \subset X$, then

- (a) $\overline{E}$ is closed,
- (b) $E = \overline{E}$ if and only if E is closed,
- (c) $\overline{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

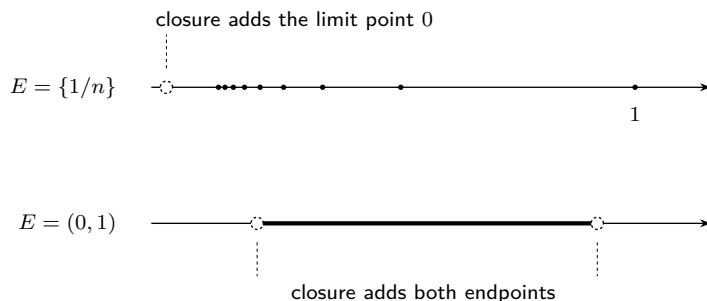
By (a) and (c), $\overline{E}$ is the smallest closed subset of X that contains E .²⁰

Proof.

- (a) If $p \in X$ and $p \notin \overline{E}$ then p is neither a point of E nor a limit point of E . Hence p has a neighborhood which does not intersect E . The complement of $\overline{E}$ is therefore open. Hence $\overline{E}$ is closed.
- (b) If $E = \overline{E}$, (a) implies that E is closed. If E is closed, then $E' \subset E$ [by Definitions 2.18(d) and 2.26], hence $E = \overline{E}$.
- (c) If F is closed and $F \supset E$, then $F \supset F'$, hence $F \supset E'$. Thus $F \supset \overline{E}$.

□

²⁰ That last sentence is the content; (a) and (b) are warm-up. “Smallest closed set containing E ” is the characterisation you should carry — it is how closure is defined in most later treatments, with $\overline{E} = E \cup E'$ demoted to a theorem.



Closure fills in exactly the limit points (dashed circles); nothing else changes.
 Compare 2.28: for a bounded set of reals, $\sup E$ is always among the points filled in, if it was missing.

2.28 Theorem. Let E be a nonempty set of real numbers which is bounded above. Let $y = \sup E$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.

Compare this with the examples in Sec. 1.9.

Proof. If $y \in E$ then $y \in \overline{E}$. Assume $y \notin E$. For every $h > 0$ there exists then a point $x \in E$ such that $y - h < x < y$, for otherwise $y - h$ would be an upper bound of E . Thus y is a limit point of E . Hence $y \in \overline{E}$. $\square$

CHAPTER 1 MEETS CHAPTER 2

This is the first theorem to connect the order structure of $\mathbb{R}$ with the topology, and the proof is two lines because the work was done in Chapter 1. “For every $h > 0$ there is $x \in E$ with $y - h < x$ ” is precisely clause (ii) of Definition 1.8, in the working form given there — and clause (ii) says exactly that y is crowded from below by E , which is what a limit point is.

So $\sup E$ is either a member of E or a limit point of it, never anything else. The examples at 1.9 now read topologically: $\sup E_1 = 0 \notin E_1$ makes 0 a limit point, while $\sup E_2 = 0 \in E_2$. Theorem 2.28 is what lets Chapter 4 conclude that a continuous function on a compact set attains its supremum (4.16).

2.29 Remark. Suppose $E \subset Y \subset X$, where X is a metric space. To say that E is an open subset of X means that to each point $p \in E$ there is associated a positive number r such that the conditions $d(p, q) < r$, $q \in X$ imply that $q \in E$. But we have already observed (Example 2.16) that Y is also a metric space, so that our definitions may equally well be made within Y . To be quite explicit, let us say that E is *open relative to Y* if to each $p \in E$ there is associated an $r > 0$ such that $q \in E$ whenever $d(p, q) < r$ and $q \in Y$. Example 2.21(g) showed that a set may be open relative to Y without being an open subset of X . However, there is a simple relation between these concepts, which we now state.

2.30 Theorem. Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X .

Proof. Suppose E is open relative to Y . To each $p \in E$ there is a positive number r_p such that the conditions $d(p, q) < r_p$, $q \in Y$ imply that $q \in E$. Let V_p be the set of all $q \in X$ such that $d(p, q) < r_p$, and define

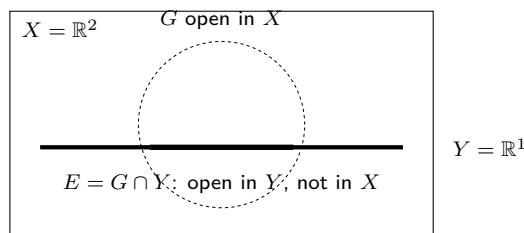
$$G = \bigcup_{p \in E} V_p.$$

Then G is an open subset of X , by Theorems 2.19 and 2.24.

Since $p \in V_p$ for all $p \in E$, it is clear that $E \subset G \cap Y$.

By our choice of V_p , we have $V_p \cap Y \subset E$ for every $p \in E$, so that $G \cap Y \subset E$. Thus $E = G \cap Y$, and one half of the theorem is proved.

Conversely, if G is open in X and $E = G \cap Y$, every $p \in E$ has a neighborhood $V_p \subset G$. Then $V_p \cap Y \subset E$, so that E is open relative to Y . $\square$



The same points form an open set inside the line and a non-open set inside the plane. “Open” always means open in something.

COMPACT SETS

THE CENTRE OF THE CHAPTER | Everything so far has been vocabulary. This section introduces the one property that does real work, and it arrives in a form that looks arbitrary: a set is compact if every open cover has a finite subcover.

Why that definition. Compactness is the hypothesis that converts “for each point there is some local fact” into “there is one fact for the whole set.” Local statements come with a neighbourhood attached; a finite subcover lets you take a minimum over finitely many of them, and a minimum of finitely many positive numbers is positive (the move already used at 2.20 and 2.24(c)). Every application in Chapter 4 — boundedness 4.15, attainment 4.16, uniform continuity 4.19 — is that one manoeuvre.

The plan of the section. 2.33–2.37 develop compactness abstractly, in any metric space. 2.38–2.40 then do the hard work of exhibiting compact sets in $\mathbb{R}^k$, culminating in “every k -cell is compact.” 2.41 collects the payoff: in $\mathbb{R}^k$, compact means closed and bounded. Note the division — 2.33–2.37 are general, 2.38–2.42 are about $\mathbb{R}^k$ and are *false* in general metric spaces.

2.31 Definition. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \bigcup_\alpha G_\alpha$.

2.32 Definition. A subset K of a metric space X is said to be *compact* if every open cover of K contains a finite subcover.

More explicitly, this requirement is that if $\{G_\alpha\}$ is an open cover of K , then there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that

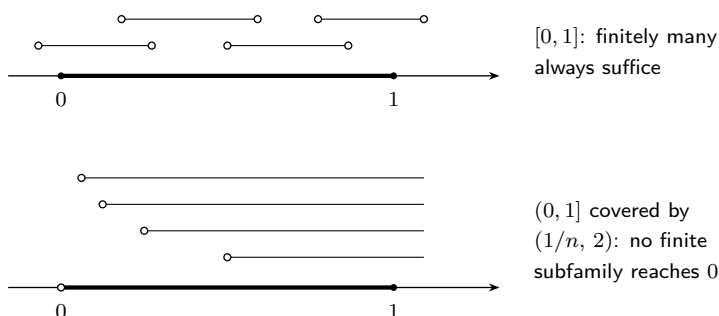
$$K \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}.$$

The notion of compactness is of great importance in analysis, especially in connection with continuity (Chap. 4).²¹

It is clear that every finite set is compact. The existence of a large class of infinite compact sets in $\mathbb{R}^k$ will follow from Theorem 2.41.

WHAT A COVER LOOKS LIKE, AND WHAT FAILS

Two examples settle the intuition. Both are covers of a subset of the line by open intervals; one collapses to finitely many, the other cannot.



Compactness fails for $(0, 1]$ because the cover creeps toward a missing endpoint. Any finite subfamily stops short.

The failure is instructive: $(0, 1]$ is bounded but not closed, and the cover exploits exactly the missing point. Theorem 2.41 will say that in $\mathbb{R}^k$ these are the only two ways compactness can fail.

We observed earlier (in Remark 2.29) that if $E \subset Y \subset X$, then E may be open relative to Y without being open relative to X . The property of being open thus depends on the space in which E is embedded. The same is true of the property of being closed.

Compactness, however, behaves better, as we shall now see. To formulate the next theorem, let us say, temporarily, that K is *compact relative to X* if the requirements of Definition 2.32 are met.

2.33 Theorem. Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y .

By virtue of this theorem we are able, in many situations, to regard compact sets as metric spaces in their own right, without paying any attention to any embedding space. In particular, although it makes little

21 Read the quantifiers carefully.

Every cover must admit a finite subcover — exhibiting one cover that happens to have a finite subcover proves nothing. To show a set is not compact it suffices to produce a single bad cover, which is why non-compactness is usually the easy direction.

22 This is why compactness is the better notion. “Open” and “closed” are relative and must always be qualified; compactness is intrinsic — a property of K with its own metric, independent of where K happens to sit. That is what makes it usable as a hypothesis in Chapter 4, where the ambient space changes from line to line.

sense to talk of open spaces, or of closed spaces (every metric space X is an open subset of itself, and is a closed subset of itself), it does make sense to talk of *compact metric spaces*.²²

Proof. Suppose K is compact relative to X , and let $\{V_\alpha\}$ be a collection of sets, open relative to Y , such that $K \subset \bigcup_\alpha V_\alpha$. By Theorem 2.30, there are sets G_α , open relative to X , such that $V_\alpha = Y \cap G_\alpha$, for all α ; and since K is compact relative to X , we have

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n} \quad (2.22)$$

for some choice of finitely many indices $\alpha_1, \dots, \alpha_n$. Since $K \subset Y$, (2.22) implies

$$K \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}. \quad (2.23)$$

This proves that K is compact relative to Y .

Conversely, suppose K is compact relative to Y , let $\{G_\alpha\}$ be a collection of open subsets of X which covers K , and put $V_\alpha = Y \cap G_\alpha$. Then (2.23) will hold for some choice of $\alpha_1, \dots, \alpha_n$; since $V_\alpha \subset G_\alpha$, (2.23) implies (2.22).

This completes the proof. $\square$

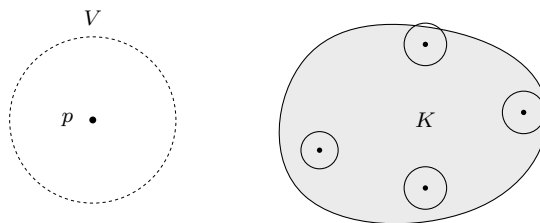
2.34 Theorem. *Compact subsets of metric spaces are closed.*

Proof. Let K be a compact subset of a metric space X . We shall prove that the complement of K is an open subset of X .

Suppose $p \in X$, $p \notin K$. If $q \in K$, let V_q and W_q be neighborhoods of p and q , respectively, of radius less than $\frac{1}{2}d(p, q)$ [see Definition 2.18(a)]. Since K is compact, there are finitely many points $q_1, \dots, q_n$ in K such that

$$K \subset W_{q_1} \cup \cdots \cup W_{q_n} = W.$$

If $V = V_{q_1} \cap \cdots \cap V_{q_n}$, then V is a neighborhood of p which does not intersect W . Hence $V \subset K^c$, so that p is an interior point of K^c . The theorem follows.²³ $\square$



W : finitely many balls covering K

23 Here is the manoeuvre the signpost promised, in its purest form. For each q we get a local separation — two small balls, radius depending on q . Compactness reduces the infinitely many q to finitely many, and only then may we intersect the V_q : an infinite intersection of neighbourhoods need not be a neighbourhood, exactly as 2.25 showed.

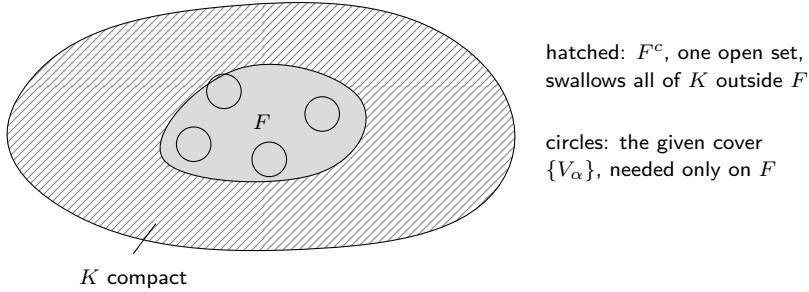
Each $q \in K$ is separated from p by a pair of balls. Finitely many W_q already cover K , so the matching finitely many V_q may be intersected to give one ball around p missing K .

24 The trick of the proof, and it is worth remembering: to convert a cover of F into a cover of the larger K , throw in the single open set F^c , which mops up everything outside F . It is legitimately open precisely because F is closed — 2.23.

2.35 Theorem. *Closed subsets of compact sets are compact.*

Proof. Suppose $F \subset K \subset X$, F is closed (relative to X), and K is compact. Let $\{V_\alpha\}$ be an open cover of F . If F^c is adjoined to $\{V_\alpha\}$,

we obtain an open cover Ω of K .²⁴ Since K is compact, there is a finite subcollection Φ of Ω which covers K , and hence F . If F^c is a member of Φ , we may remove it from Φ and still retain an open cover of F . We have thus shown that a finite subcollection of $\{V_\alpha\}$ covers F . $\square$



Adjoining the single open set F^c turns a cover of F into a cover of K ; compactness of K then hands back finitely many sets, and discarding F^c leaves a finite cover of F .

Corollary. *If F is closed and K is compact, then $F \cap K$ is compact.*

Proof. Theorems 2.24(b) and 2.34 show that $F \cap K$ is closed; since $F \cap K \subset K$, Theorem 2.35 shows that $F \cap K$ is compact. $\square$

2.36 Theorem. *If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty.*

Proof. Fix a member K_1 of $\{K_\alpha\}$ and put $G_\alpha = K_\alpha^c$. Assume that no point of K_1 belongs to every K_α . Then the sets G_α form an open cover of K_1 ; since K_1 is compact, there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that $K_1 \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$. But this means that

$$K_1 \cap K_{\alpha_1} \cap \dots \cap K_{\alpha_n}$$

is empty, in contradiction to our hypothesis. $\square$

Corollary. *If $\{K_n\}$ is a countable collection of nonempty compact sets such that $K_n \supset K_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} K_n$ is not empty.*

The finite intersection property is compactness turned inside out

2.36 is 2.32 with complements taken, and nothing more. “Every open cover has a finite subcover” says: if no finite subfamily of $\{G_\alpha\}$ covers K , then $\{G_\alpha\}$ does not cover K . Complementing every G turns “covers” into “has empty intersection with,” and the statement becomes 2.36.

The corollary is the form that gets used, and it answers the worry raised at Example 2.10(b)(iii), where nested nonempty sets had empty intersection. There the sets were the segments $(0, x)$ — bounded but not closed, hence not compact. Compactness is exactly what forbids the nesting from escaping to nothing.

Compare Theorem 2.38, the same statement for closed intervals

proved directly from the least-upper-bound property, and Yu Pin's nested interval axiom in Chapter 1. Three routes to one fact, and Rudin uses whichever is cheapest at the point of need.

2.37 Theorem. *If E is an infinite subset of a compact set K , then E has a limit point in K .*

Proof. If no point of K were a limit point of E , then each $q \in K$ would have a neighborhood V_q which contains at most one point of E (namely, q , if $q \in E$). It is clear that no finite subcollection of $\{V_q\}$ can cover E ; and the same is true of K , since $E \subset K$. This contradicts the compactness of K . $\square$

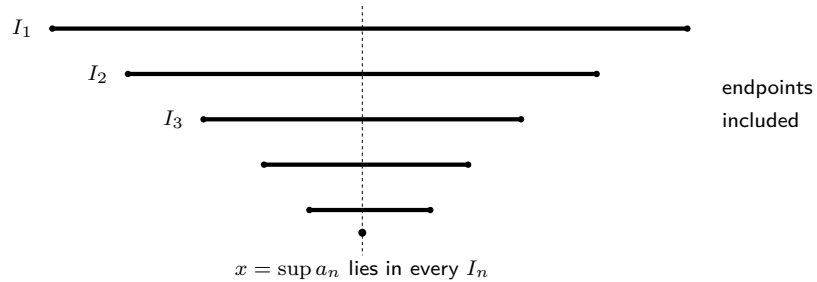
2.38 Theorem. *If $\{I_n\}$ is a sequence of intervals in $\mathbb{R}^1$, such that $I_n \supset I_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. If $I_n = [a_n, b_n]$, let E be the set of all a_n . Then E is nonempty and bounded above (by b_1). Let x be the sup of E . If m and n are positive integers, then

$$a_n \leq a_{m+n} \leq b_{m+n} \leq b_m,$$

so that $x \leq b_m$ for each m . Since it is obvious that $a_m \leq x$, we see that $x \in I_m$ for $m = 1, 2, 3, \dots$.²⁵ $\square$

25 This is Chapter 1 doing the work again — the whole proof is “take a supremum,” which exists by 1.19. Every b_m is an upper bound for the left endpoints, so the least upper bound x sits below all of them and above all the a 's. Note the closedness of the intervals is essential: for $I_n = (0, 1/n)$ the same construction gives $x = 0$, which lies in no I_n .



Left endpoints increase, right endpoints decrease, and $\sup a_n$ is caught between.
Drop the endpoints and the common point can escape.

2.39 Theorem. *Let k be a positive integer. If $\{I_n\}$ is a sequence of k -cells such that $I_n \supset I_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. Let I_n consist of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that

$$a_{n,j} \leq x_j \leq b_{n,j} \quad (1 \leq j \leq k; n = 1, 2, 3, \dots),$$

and put $I_{n,j} = [a_{n,j}, b_{n,j}]$. For each j , the collection $\{I_{n,j}\}$ satisfies the hypotheses of Theorem 2.38. Hence there are real numbers x_j^* ($1 \leq j \leq k$) such that

$$a_{n,j} \leq x_j^* \leq b_{n,j} \quad (1 \leq j \leq k; n = 1, 2, 3, \dots).$$

Setting $\mathbf{x}^* = (x_1^*, \dots, x_k^*)$, we see that $\mathbf{x}^* \in I_n$ for $n = 1, 2, 3, \dots$. The theorem follows.²⁶ $\square$

26 The k -dimensional statement is k separate one-dimensional statements, because a k -cell is a product of intervals and a point lies in it exactly when each coordinate lies in the corresponding interval. This coordinatewise reduction is used again in 2.40 and throughout Chapter 9.

2.40 Theorem. *Every k -cell is compact.*

Proof. Let I be a k -cell, consisting of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that $a_j \leq x_j \leq b_j$ ($1 \leq j \leq k$). Put

$$\delta = \left\{ \sum_1^k (b_j - a_j)^2 \right\}^{1/2}.$$

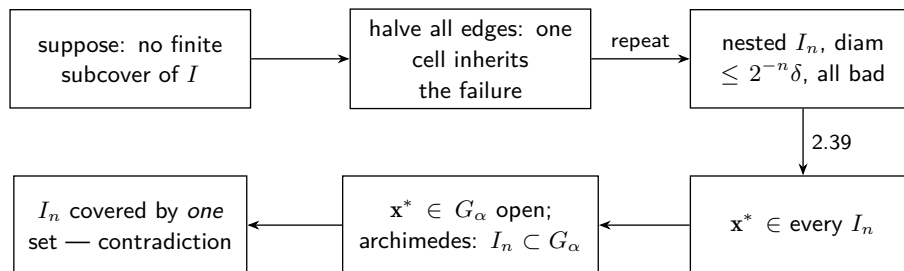
Then $|\mathbf{x} - \mathbf{y}| \leq \delta$, if $\mathbf{x} \in I$, $\mathbf{y} \in I$.

Suppose, to get a contradiction, that there exists an open cover $\{G_\alpha\}$ of I which contains no finite subcover of I . Put $c_j = (a_j + b_j)/2$. The intervals $[a_j, c_j]$ and $[c_j, b_j]$ then determine 2^k k -cells Q_i whose union is I . At least one of these sets Q_i , call it I_1 , cannot be covered by any finite subcollection of $\{G_\alpha\}$ (otherwise I could be so covered). We next subdivide I_1 and continue the process. We obtain a sequence (I_n) with the following properties:

- (a) $I \supset I_1 \supset I_2 \supset I_3 \supset \dots$;
- (b) I_n is not covered by any finite subcollection of $\{G_\alpha\}$;
- (c) If $\mathbf{x} \in I_n$ and $\mathbf{y} \in I_n$, then $|\mathbf{x} - \mathbf{y}| \leq 2^{-n}\delta$.

By (a) and Theorem 2.39, there is a point $\mathbf{x}^*$ which lies in every I_n . For some α , $\mathbf{x}^* \in G_\alpha$. Since G_α is open, there exists $r > 0$ such that $|\mathbf{y} - \mathbf{x}^*| < r$ implies that $\mathbf{y} \in G_\alpha$. If n is so large that $2^{-n}\delta < r$ (there is such an n , for otherwise $2^n \leq \delta/r$ for all positive integers n , which is absurd since $\mathbb{R}$ is archimedean), then (c) implies that $I_n \subset G_\alpha$, which contradicts (b).

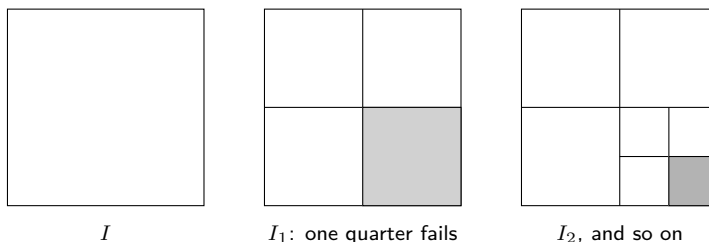
This completes the proof. $\square$



The proof as a flow diagram. Completeness enters once (2.39, via 2.38), the archimedean property once, and the contradiction is that one is finite.

BISECTION: THE SHAPE OF THE ARGUMENT

This is the most important proof in the chapter, and its structure recurs throughout analysis. Assume the conclusion fails, then manufacture a nested sequence of ever-smaller pieces in which it keeps failing, and squeeze them onto a single point where the failure becomes absurd.



Halving each edge gives 2^k pieces; at least one inherits the failure. Diameters fall like $2^{-n}\delta$, so the nested pieces close on a point.

Three ingredients, each traceable to Chapter 1.

Something survives the nesting — Theorem 2.39, which rests on 2.38, which rests on the least-upper-bound property. This is where completeness enters, and it is the only place. Over $\mathbb{Q}$ the argument breaks here and k -cells are not compact.

The pieces shrink to nothing — property (c). Diameters are halved each time.

Shrinking beats any fixed radius — the archimedean property, which Rudin flags explicitly in parentheses. Without it $2^{-n}\delta$ might stay above r forever.

Then the contradiction is immediate: I_n fits inside the single set G_α , so one set covers it, contradicting “no finite subcollection covers I_n .” One is finite.

Seen from the comparison at Definition 1.10: this proof *is* the implication “nested intervals $\Rightarrow$ Heine–Borel” from the ECNU text’s cycle of six equivalent completeness statements, specialised to cells. Bisection is the standard bridge between the two.

The equivalence of (a) and (b) in the next theorem is known as the *Heine–Borel theorem*.

2.41 Theorem. *If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two:*

- (a) *E is closed and bounded.*
- (b) *E is compact.*
- (c) *Every infinite subset of E has a limit point in E .*

Proof. If (a) holds, then $E \subset I$ for some k -cell I , and (b) follows from Theorems 2.40 and 2.35. Theorem 2.37 shows that (b) implies (c). It remains to be shown that (c) implies (a).

If E is not bounded, then E contains points $\mathbf{x}_n$ with

$$|\mathbf{x}_n| > n \quad (n = 1, 2, 3, \dots).$$

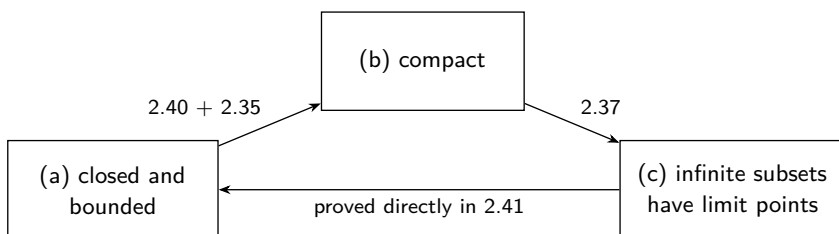
The set S consisting of these points $\mathbf{x}_n$ is infinite and clearly has no limit point in $\mathbb{R}^k$, hence has none in E . Thus (c) implies that E is bounded.

If E is not closed, then there is a point $\mathbf{x}_0 \in \mathbb{R}^k$ which is a limit point of E but not a point of E . For $n = 1, 2, 3, \dots$, there are points $\mathbf{x}_n \in E$ such that $|\mathbf{x}_n - \mathbf{x}_0| < 1/n$. Let S be the set of these points $\mathbf{x}_n$. Then S is infinite (otherwise $|\mathbf{x}_n - \mathbf{x}_0|$ would have a constant positive value, for infinitely many n), S has $\mathbf{x}_0$ as a limit point, and S has no other limit point in $\mathbb{R}^k$. For if $\mathbf{y} \neq \mathbf{x}_0$, then

$$\begin{aligned} |\mathbf{x}_n - \mathbf{y}| &\geq |\mathbf{x}_0 - \mathbf{y}| - |\mathbf{x}_n - \mathbf{x}_0| \\ &\geq |\mathbf{x}_0 - \mathbf{y}| - \frac{1}{n} \geq \frac{1}{2}|\mathbf{x}_0 - \mathbf{y}| \end{aligned}$$

for all but finitely many n ; this shows that $\mathbf{y}$ is not a limit point of S (Theorem 2.20). Thus S has no limit point in E ; hence E must be closed if (c) holds. $\square$

We should remark, at this point, that (b) and (c) are equivalent in any metric space (Exercise 2.26), but that (a) does not, in general, imply (b) and (c). Examples are furnished by Exercise 2.16 and by the space $\mathcal{L}^2$, which is discussed in Chap. 11.



The cycle proved in 2.41. In a general metric space the arrow (a)→(b) is the one that breaks (Exercise 2.16, $\mathcal{L}^2$), while (b)↔(c) survives everywhere (2.37 and Exercise 2.26).

Where this sits among the completeness theorems

Rudin's remark deserves emphasis: **(a) is the odd one out.** "Closed and bounded" is not equivalent to compactness in a general metric space — it is a fact about $\mathbb{R}^k$ specifically, and the whole apparatus 2.38–2.40 exists to establish it. Exercise 2.16 gives a closed bounded non-compact set inside $\mathbb{Q}$; in $\mathcal{L}^2$ the closed unit ball fails too.

This is also where Chapter 2 rejoins the discussion at Definition 1.10. Of the six equivalent formulations of completeness collected in the ECNU text, three appear in this chapter: **Heine–Borel** is (a)↔(b) here, **Bolzano–Weierstrass** is 2.42, and the **nested interval** property is 2.38. Rudin proves each separately from the least-upper-bound property rather than deriving them from one another, so their equivalence never becomes visible. It is worth knowing that the four statements — least-upper-bound, nested intervals, Heine–Borel, Bolzano–Weierstrass — are interderivable, and that Rudin's ordering is a choice, not a necessity.

Tao (*Analysis II*, 1.5) takes the opposite route: he defines compactness by sequences (every sequence has a convergent subsequence, Rudin's (c)) and proves the open-cover version afterwards. That ordering is friendlier, and it is why (c) is often the easier property to verify in practice — but it does not generalise past metric spaces, whereas 2.32 does.

2.42 Theorem (Weierstrass). *Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.*

Proof. Being bounded, the set E in question is a subset of a k -cell $I \subset \mathbb{R}^k$. By Theorem 2.40, I is compact, and so E has a limit point in I , by Theorem 2.37. $\square$

PERFECT SETS

TWO RESULTS, ONE OF THEM A WARNING | Theorem 2.43 is a second, self-contained proof that $\mathbb{R}$ is uncountable, using compactness instead of Cantor's diagonal — and unlike 2.14 it needs no discussion of binary expansions.

Section 2.44 is the real business. The Cantor set is the standing counterexample of the subject: compact, perfect, uncountable, yet containing no interval and (as Chapter 11 will show) of measure zero. Every intuition of the form “a set that big must contain an interval” dies here. Expect to meet it again in Exercises 2.17–2.18, in Exercise 6.6 (integration against the Cantor set), and in Chapter 11.

2.43 Theorem. *Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is uncountable.*

Proof. Since P has limit points, P must be infinite. Suppose P is countable, and denote the points of P by $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots$. We shall construct a countable collection $\{V_n\}$ of neighborhoods, as follows.²⁷

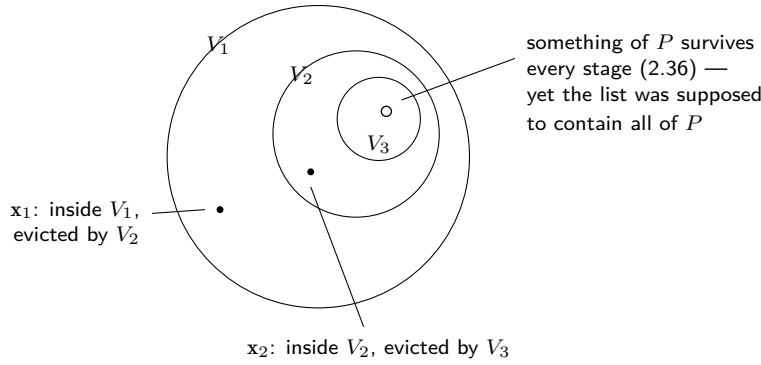
Let V_1 be any neighborhood of $\mathbf{x}_1$. If V_1 consists of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}_1| < r$, then the closure $\overline{V}_1$ of V_1 is the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}_1| \leq r$.

Suppose V_n has been constructed, so that $V_n \cap P$ is not empty. Since every point of P is a limit point of P , there is a neighborhood V_{n+1} such that (i) $\overline{V}_{n+1} \subset V_n$, (ii) $\mathbf{x}_n \notin \overline{V}_{n+1}$, (iii) $V_{n+1} \cap P$ is not empty.²⁸ By (iii), V_{n+1} satisfies our induction hypothesis, and the construction can proceed.

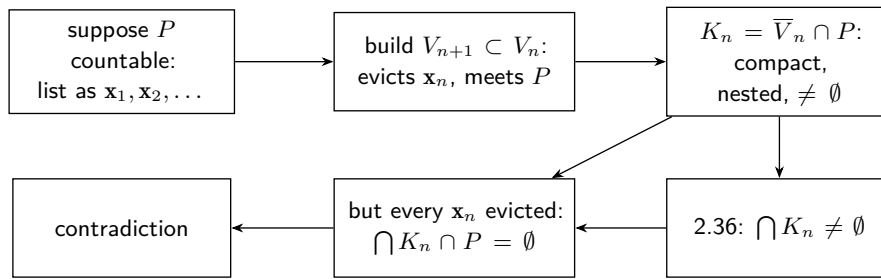
Put $K_n = \overline{V}_n \cap P$. Since $\overline{V}_n$ is closed and bounded, $\overline{V}_n$ is compact. Since $\mathbf{x}_n \notin K_{n+1}$, no point of P lies in $\bigcap_{n=1}^{\infty} K_n$. Since $K_n \subset P$, this implies that $\bigcap_{n=1}^{\infty} K_n$ is empty. But each K_n is nonempty, by (iii), and $K_n \supset K_{n+1}$, by (i); this contradicts the Corollary to Theorem 2.36. $\square$

²⁷ The strategy is Cantor's diagonal in disguise: assume a list of all of P , then trap a point that the list misses. Where 2.14 built the missing object digit by digit, here it is caught by a nested sequence of compact sets — and 2.36 guarantees something survives.

²⁸ Clause (ii) is the diagonal step — it evicts the n th listed point at stage n . Clause (iii) is where perfect is used: because $\mathbf{x}$ is a limit point of P , there are always more points of P nearby to keep the construction alive.



Stage $n+1$ shrinks inside stage n and evicts the n th listed point; perfectness keeps each stage nonempty. Compactness forces a survivor, which the list therefore missed — Cantor's diagonal, drawn with balls instead of digits.



The flow: the list drives the construction, perfectness keeps every stage alive, and compactness (2.36) collides with total eviction.

Corollary. Every interval $[a, b]$ ($a < b$) is uncountable. In particular, the set of all real numbers is uncountable.

2.44 The Cantor Set. The set which we are now going to construct shows that there exist perfect sets in $\mathbb{R}^1$ which contain no segment.

Let E_0 be the interval $[0, 1]$. Remove the segment $(\frac{1}{3}, \frac{2}{3})$, and let E_1 be the union of the intervals

$$\left[0, \frac{1}{3}\right], \quad \left[\frac{2}{3}, 1\right].$$

Remove the middle thirds of these intervals, and let E_2 be the union of the intervals

$$\left[0, \frac{1}{9}\right], \quad \left[\frac{2}{9}, \frac{3}{9}\right], \quad \left[\frac{6}{9}, \frac{7}{9}\right], \quad \left[\frac{8}{9}, 1\right].$$

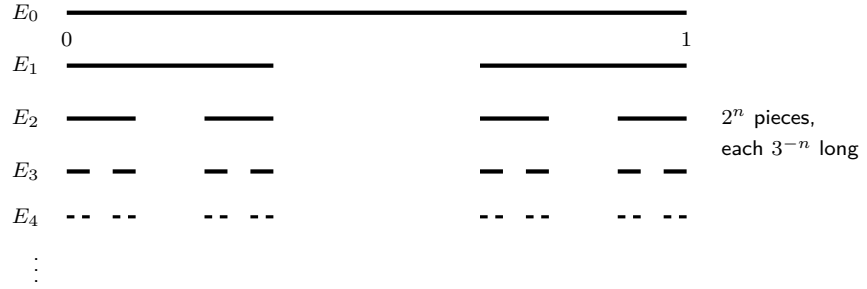
Continuing in this way, we obtain a sequence of compact sets E_n such that

- (a) $E_1 \supset E_2 \supset E_3 \supset \cdots$;
- (b) E_n is the union of 2^n intervals, each of length 3^{-n} .

The set

$$P = \bigcap_{n=1}^{\infty} E_n$$

is called the *Cantor set*. P is clearly compact, and Theorem 2.36 shows that P is not empty.



Removing middle thirds. Total length removed is $\sum 2^{n-1}3^{-n} = 1$, so “nothing” is left — yet what remains is uncountable.

No segment of the form

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right) \quad (2.24)$$

where k and m are positive integers, has a point in common with P . Since every segment (α, β) contains a segment of the form (2.24), if

$$3^{-m} < \frac{\beta - \alpha}{6},$$

P contains no segment.

To show that P is perfect, it is enough to show that P contains no isolated point. Let $x \in P$, and let S be any segment containing x . Let I_n be that interval of E_n which contains x . Choose n large enough, so that $I_n \subset S$. Let x_n be an endpoint of I_n , such that $x_n \neq x$.

It follows from the construction of P that $x_n \in P$. Hence x is a limit point of P , and P is perfect.²⁹

One of the most interesting properties of the Cantor set is that it provides us with an example of an uncountable set of measure zero (the concept of measure will be discussed in Chap. 11).

Everything the Cantor set breaks

It is worth listing what P demonstrates, because each item contradicts a plausible belief.

Uncountable but containing no interval. By 2.43 it is uncountable, being perfect and nonempty; yet it contains no segment. So “uncountable” carries no implication of thickness.

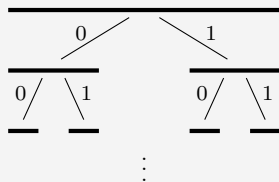
Total length zero. The removed intervals have total length $\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = 1$. What remains has measure zero (Chapter 11) — an uncountable set of measure zero.

Every point is a limit of *other* points, yet no point has neighbours in bulk. P is perfect, so no isolated points; but it is also nowhere dense, so its closure has empty interior. Both at once.

It is not a curiosity. P is homeomorphic to $\{0, 1\}^{\mathbb{N}}$, which is why 2.14 and 2.43 measure the same thing — a point of P is exactly a choice of “left third or right third” at every stage, i.e. a 0–1 sequence. That correspondence also makes the uncountability

²⁹ The endpoints are the key: they are never removed at any later stage, because removal always takes an open middle third. So every $x \in P$ has other points of P arbitrarily close — the endpoints of the shrinking interval containing it.

immediate without 2.43.



a point of P = an infinite
path = its address, e.g. $0, 1, 1, 0, \dots$
— exactly a 0–1 sequence (2.14)

*Left third or right third, forever: the surviving intervals form a binary tree,
and a point of P is an infinite path through it.*

CONNECTED SETS

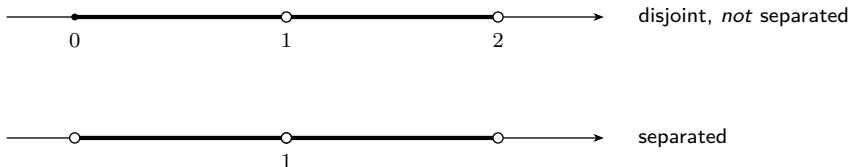
TWO DEFINITIONS AND ONE THEOREM | The shortest section in the chapter, and it exists for the intermediate value theorem (4.23), which will follow in one line from Theorem 4.22: continuous images of connected sets are connected.

The only subtlety is that “connected” is defined via *separated* sets, not disjoint ones. Remark 2.46 is there to stop you conflating the two, and 2.47 shows that on the line the whole notion collapses to something familiar: connected means interval.

2.45 Definition. Two subsets A and B of a metric space X are said to be *separated* if both $A \cap \overline{B}$ and $\overline{A} \cap B$ are empty, i.e., if no point of A lies in the closure of B and no point of B lies in the closure of A .

A set $E \subset X$ is said to be *connected* if E is not a union of two nonempty separated sets.

2.46 Remark. Separated sets are of course disjoint, but disjoint sets need not be separated. For example, the interval $[0, 1]$ and the segment $(1, 2)$ are not separated, since 1 is a limit point of $(1, 2)$. However, the segments $(0, 1)$ and $(1, 2)$ are separated.



*The difference is whether the shared boundary point belongs to one of the sets.
Closures must miss each other, not merely the sets themselves.*

SEPARATED IS STRONGER THAN DISJOINT, AND THE GAP MATTERS

If “connected” were defined by disjointness, *every* set with more than one point would be disconnected — split $[0, 1]$ into $[0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ and you have two disjoint nonempty pieces. The definition would be useless.

Requiring the *closures* to miss is what rules this out: $\frac{1}{2}$ lies in the closure of the second piece, so the split is not a separation. In effect, separated sets are ones that do not touch even in the limit.

The connected subsets of the line have a particularly simple structure:

2.47 Theorem. *A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property: If $x \in E$, $y \in E$, and $x < z < y$, then $z \in E$.*

Proof. If there exist $x \in E$, $y \in E$, and some $z \in (x, y)$ such that $z \notin E$, then $E = A_z \cup B_z$ where

$$A_z = E \cap (-\infty, z), \quad B_z = E \cap (z, \infty).$$

Since $x \in A_z$ and $y \in B_z$, A and B are nonempty. Since $A_z \subset (-\infty, z)$ and $B_z \subset (z, \infty)$, they are separated. Hence E is not connected.

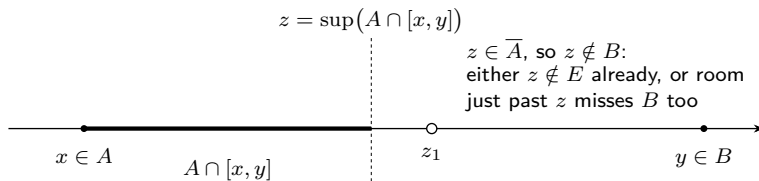
To prove the converse, suppose E is not connected. Then there are nonempty separated sets A and B such that $A \cup B = E$. Pick $x \in A$, $y \in B$, and assume (without loss of generality) that $x < y$. Define

$$z = \sup(A \cap [x, y]).$$

By Theorem 2.28, $z \in \bar{A}$; hence $z \notin B$. In particular, $x \leq z < y$.³⁰

If $z \notin A$, it follows that $x < z < y$ and $z \notin E$.

If $z \in A$, then $z \notin \bar{B}$, hence there exists z_1 such that $z < z_1 < y$ and $z_1 \notin B$. Then $x < z_1 < y$ and $z_1 \notin E$. $\square$



The same seam picture as Definition 1.8, one chapter later: the supremum lands in $\bar{A}$ and hence off B , and a point missing from E is found at or just past the seam.

CONNECTED ON THE LINE MEANS INTERVAL

The property in 2.47 — if two points are in E , everything between them is too — is exactly the definition of an interval, in the general sense that includes $\emptyset$, single points, half-lines and $\mathbb{R}$ itself. So the theorem reads: *the connected subsets of $\mathbb{R}$ are precisely the intervals*.

That is worth holding because it makes the intermediate value theorem (4.23) transparent. If f is continuous on $[a, b]$ then $f([a, b])$ is connected (4.22), hence an interval by 2.47, hence contains every value between $f(a)$ and $f(b)$ — the intermediate value theorem. Chapter 1's Theorem 1.21 on n th roots was a hand-rolled special case of exactly this, as the note there observed.

³⁰ Chapter 1 again. The supremum exists by 1.19, lands in $\bar{A}$ by 2.28, and therefore avoids B because A and B are separated — that is the one place separation is spent. Every step of this converse is an application of something already proved.

Note also that connectedness, unlike openness, is intrinsic: like compactness it survives change of ambient space, since 2.45 refers only to closures within E 's own metric.

CHAPTER 2 IN ONE PARAGRAPH

Countability sorts infinite sets into two sizes, and $\mathbb{Q}$ and $\mathbb{R}$ land on opposite sides (2.13, 2.43). Metric spaces distill $\mathbb{R}^k$ down to its distance function, and open, closed and limit point are the vocabulary that distance alone supports (2.15–2.30). Compactness — every open cover admits a finite subcover — is the property that turns local facts into global ones, and in $\mathbb{R}^k$ it is exactly closed-and-bounded (2.41), a fact that rests on completeness through the nested-interval theorems (2.38–2.40) and on nothing else. Perfect sets are uncountable (2.43), and the Cantor set shows how thin one can be (2.44). Connected subsets of the line are precisely the intervals (2.47). Chapter 4 will spend all of it: continuity gets defined by inverse images of open sets, compactness becomes boundedness and attainment of extrema, and connectedness becomes the intermediate value theorem.

EXERCISES

HOW TO USE THESE

Chapter 2's exercises are less optional than Chapter 1's. Several supply results the book later assumes, and one chain (2.22–2.26) proves a theorem Rudin states in the text without proof.

The chain that matters. $2.22 \rightarrow 2.23 \rightarrow 2.24 \rightarrow 2.26$ builds up to the claim made in passing after 2.41: that (b) and (c) are equivalent in *any* metric space, not just $\mathbb{R}^k$. Rudin proves $(b) \Rightarrow (c)$ as 2.37 and leaves the converse here. Do them in order; each is a step.

Counterexample exercises. 2.14, 2.15, 2.16 show what fails without compactness, and 2.16 is the one Rudin cites in the text — a closed bounded set that is not compact, living in $\mathbb{Q}$.

Baire. 2.30 is a special case of Baire's theorem, proved by imitating 2.43. It is used throughout functional analysis, and Exercise 3.22 gives the general version.

Routine. 2.1, 2.5–2.9, 2.11 are short checks on the definitions. Do them quickly to make sure the vocabulary of 2.18 has landed.

2.1. Prove that the empty set is a subset of every set.³¹

³¹ *Vacuous implication:* “every $x \in \emptyset$ satisfies . . .” is true because there is no x to check. If you find this uncomfortable, note the alternative would make $\emptyset \subset \emptyset$ false.

32 The hint is the whole exercise, and it is the anti-diagonal idea of 2.12 again: sort the objects by a size parameter N so that each level is finite, then take a countable union. Each equation has at most n roots, so level N contributes finitely many numbers.

33 One line from 2.2 and 2.43: the algebraic numbers are countable, $\mathbb{R}$ is not, so something is left over. Note that this proves transcendentals exist without exhibiting a single one — Cantor's original observation, and a century younger than the first explicit example.

34 No, and the argument is the same as 2.3: if it were, $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be a union of two countable sets, hence countable by 2.12.

35 Take three copies of $\{1/n\}$, translated. The point is that limit points can be prescribed at will — there is nothing canonical about them.

36 The last question is the interesting one and the answer is no. Take $E = \{1/n\}$: then $E' = \{0\}$, which has no limit points at all, while E has one. Adding limit points can destroy them.

37 Finite versus infinite once more — (a) is an equality, (b) only an inclusion. For the proper inclusion enumerate $\mathbb{Q}$ as $A_i = \{q_i\}$: each $\overline{A_i}$ is a point, their union is $\mathbb{Q}$, but $\overline{B} = \mathbb{R}$.

38 Yes for open, no for closed. An open set contains a ball about each of its points, and a ball in $\mathbb{R}^2$ has infinitely many points; a closed set may be a single point. So open sets in $\mathbb{R}^k$ ($k \geq 1$) never have isolated points — which is why the perfect sets of 2.44 are interesting: they are closed with no isolated points either, without containing any ball.

39 Parts (a)–(c) are 2.27 with every inclusion reversed: interior is the largest open set inside, as closure is the smallest closed set outside. Part (d), $(E^\circ)^c = \overline{E^c}$, is the duality that makes the two notions one notion. For (e) and (f) the answer is no both times; $E = \mathbb{Q}$ settles both.

2.2. A complex number z is said to be *algebraic* if there are integers $a_0, \dots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0.$$

Prove that the set of all algebraic numbers is countable. *Hint:* For every positive integer N there are only finitely many equations with

$$n + |a_0| + |a_1| + \dots + |a_n| = N.$$

32

2.3. Prove that there exist real numbers which are not algebraic.³³

2.4. Is the set of all irrational real numbers countable?³⁴

2.5. Construct a bounded set of real numbers with exactly three limit points.³⁵

2.6. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?³⁶

2.7. Let $A_1, A_2, A_3, \dots$ be subsets of a metric space.

(a) If $B_n = \bigcup_{i=1}^n A_i$, prove that $\overline{B_n} = \bigcup_{i=1}^n \overline{A_i}$ for $n = 1, 2, 3, \dots$

(b) If $B = \bigcup_{i=1}^\infty A_i$, prove that $\overline{B} \supset \bigcup_{i=1}^\infty \overline{A_i}$.

Show, by an example, that this inclusion can be proper.³⁷

2.8. Is every point of every open set $E \subset \mathbb{R}^2$ a limit point of E ? Answer the same question for closed sets in $\mathbb{R}^2$.³⁸

2.9. Let E° denote the set of all interior points of a set E . [See Definition 2.18(e); E° is called the *interior* of E .]

(a) Prove that E° is always open.

(b) Prove that E is open if and only if $E^\circ = E$.

(c) If $G \subset E$ and G is open, prove that $G \subset E^\circ$.

(d) Prove that the complement of E° is the closure of the complement of E .

(e) Do E and $\overline{E}$ always have the same interiors?

(f) Do E and E° always have the same closures?

39

2.10. Let X be an infinite set. For $p \in X$ and $q \in X$, define

$$d(p, q) = \begin{cases} 1 & (\text{if } p \neq q) \\ 0 & (\text{if } p = q). \end{cases}$$

Prove that this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?⁴⁰

2.11. For $x \in \mathbb{R}^1$ and $y \in \mathbb{R}^1$, define

$$\begin{aligned} d_1(x, y) &= (x - y)^2, & d_2(x, y) &= \sqrt{|x - y|}, \\ d_3(x, y) &= |x^2 - y^2|, & d_4(x, y) &= |x - 2y|, \\ d_5(x, y) &= \frac{|x - y|}{1 + |x - y|}. \end{aligned}$$

Determine, for each of these, whether it is a metric or not.⁴¹

2.12. Let $K \subset \mathbb{R}^1$ consist of 0 and the numbers $1/n$ for $n = 1, 2, 3, \dots$. Prove that K is compact directly from the definition (without using the Heine-Borel theorem).⁴²

2.13. Construct a compact set of real numbers whose limit points form a countable set.⁴³

2.14. Give an example of an open cover of the segment $(0, 1)$ which has no finite subcover.⁴⁴

2.15. Show that Theorem 2.36 and its Corollary become false (in $\mathbb{R}^1$, for example) if the word “compact” is replaced by “closed” or by “bounded.”⁴⁵

2.16. Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but that E is not compact. Is E open in $\mathbb{Q}$?⁴⁶

2.17. Let E be the set of all $x \in [0, 1]$ whose decimal expansion contains only the digits 4 and 7. Is E countable? Is E dense in $[0, 1]$? Is E compact? Is E perfect?⁴⁷

2.18. Is there a nonempty perfect set in $\mathbb{R}^1$ which contains no rational number?⁴⁸

2.19. (a) If A and B are disjoint closed sets in some metric space X , prove that they are separated.

(b) Prove the same for disjoint open sets.

[40] The discrete metric, and the standard sanity check on every definition in the chapter. Every subset is open (the ball of radius $\frac{1}{2}$ about p is $\{p\}$), hence every subset is closed, and the compact sets are exactly the finite ones — cover an infinite set by singletons and no finite subfamily suffices. Note the consequence: here every set is closed and bounded, yet almost none is compact. That is Rudin’s remark after 2.41, in its cheapest form.

41 Check the three axioms of 2.15 one at a time and the failures are quick: d_1 fails the triangle inequality, d_3 fails $d(p, q) > 0$ for $p \neq q$ (take $x = -y$), d_4 fails symmetry and even $d(p, p) = 0$. The two that work, d_2 and d_5 , are worth remembering — d_5 is the standard way to make any metric bounded without changing which sets are open.

42 The model example, and worth doing by hand. Whatever cover is given, some G_α contains 0; being open it contains all $1/n$ beyond some index, leaving only finitely many points to cover one at a time. Compare 2.32: this is the compactness proof in miniature, and it is why $\{1/n\}$ alone — without 0 — is not compact.

43 Nest the previous example: put a shrunk copy of K near each $1/n$. The limit points are then $\{1/n\} \cup \{0\}$.

44 $\{(1/n, 1)\}$. See the figure at 2.32 — this is exactly the picture drawn there.

45 For “closed” take $F_n = [n, \infty)$; for “bounded” take $(0, 1/n)$. Both are nested with empty intersection, and each fails exactly one half of “closed and bounded.” Compare Example 2.10(b)(iii).

46 The exercise Rudin cites in the text after 2.41, and the cleanest demonstration that (a) does not imply (b) outside $\mathbb{R}^k$. E is closed in $\mathbb{Q}$ because its would-be boundary points $\pm\sqrt{2}$, $\pm\sqrt{3}$ are not rational — the holes in $\mathbb{Q}$ from Chapter 1 are doing the work. It is also open in $\mathbb{Q}$, so here is a set that is both, in a space that is connected nowhere. Everything that goes wrong traces back to $\mathbb{Q}$ lacking the least-upper-bound property.

47 A Cantor set in disguise: uncountable (diagonalise, 2.14), nowhere dense, compact and perfect. Build it as an intersection of nested closed sets exactly as in 2.44 and every answer follows from that section.

48 Yes. Repeat the construction of 2.44 but at stage n remove an open interval around the n th rational, small enough to leave two closed pieces. Enumerating $\mathbb{Q}$ is possible by the Corollary to 2.13 — this is one of the places countability is actually used.

49 (a) and (b) say that for closed or open sets, disjoint already implies separated — the distinction Remark 2.46 draws only bites for sets that are neither. (d) is the striking one: connectedness forces uncountability. By (c) the space splits at every radius δ unless some point sits at distance exactly δ from p ; so all but countably many δ would disconnect X , and there must be uncountably many distances.

50 Closures yes, interiors no — and the asymmetry is easy to miss. For the counterexample take two closed discs in the plane touching at a single point: the union is connected, but removing the boundary disconnects the interior into two open discs.

51 The strategy: pull a question about $\mathbb{R}^k$ back to a question about the line, where 2.47 settles everything. Part (b) is 2.47 applied to $A_0 \cup B_0$, which cannot be all of $[0, 1]$ because that would be connected. This is the first appearance of “restrict to a segment,” a device used constantly in Chapter 9.

(c) Fix $p \in X$, $\delta > 0$, define A to be the set of all $q \in X$ for which $d(p, q) < \delta$, define B similarly, with $>$ in place of $<$. Prove that A and B are separated.

(d) Prove that every connected metric space with at least two points is uncountable. *Hint:* Use (c).

49

2.20. Are closures and interiors of connected sets always connected? (Look at subsets of $\mathbb{R}^2$.)⁵⁰

2.21. Let A and B be separated subsets of some $\mathbb{R}^k$, suppose $\mathbf{a} \in A$, $\mathbf{b} \in B$, and define

$$\mathbf{p}(t) = (1 - t)\mathbf{a} + t\mathbf{b}$$

for $t \in \mathbb{R}^1$. Put $A_0 = \mathbf{p}^{-1}(A)$, $B_0 = \mathbf{p}^{-1}(B)$. [Thus $t \in A_0$ if and only if $\mathbf{p}(t) \in A$.]

(a) Prove that A_0 and B_0 are separated subsets of $\mathbb{R}^1$.

(b) Prove that there exists $t_0 \in (0, 1)$ such that $\mathbf{p}(t_0) \notin A \cup B$.

(c) Prove that every convex subset of $\mathbb{R}^k$ is connected.

51

2.22. A metric space is called *separable* if it contains a countable dense subset. Show that $\mathbb{R}^k$ is separable. *Hint:* Consider the set of points which have only rational coordinates.⁵²

2.23. A collection $\{V_\alpha\}$ of open subsets of X is said to be a *base* for X if the following is true: For every $x \in X$ and every open set $G \subset X$ such that $x \in G$, we have $x \in V_\alpha \subset G$ for some α . In other words, every open set in X is the union of a subcollection of $\{V_\alpha\}$.

Prove that every separable metric space has a countable base. *Hint:* Take all neighborhoods with rational radius and center in some countable dense subset of X .⁵³

2.24. Let X be a metric space in which every infinite subset has a limit point. Prove that X is separable. *Hint:* Fix $\delta > 0$, and pick $x_1 \in X$. Having chosen $x_1, \dots, x_j \in X$, choose $x_{j+1} \in X$, if possible, so that $d(x_i, x_{j+1}) \geq \delta$ for $i = 1, \dots, j$. Show that this process must stop after a finite number of steps, and that X can therefore be covered by finitely many neighborhoods of radius δ . Take $\delta = 1/n$ ($n = 1, 2, 3, \dots$), and consider the centers of the corresponding neighborhoods.⁵⁴

2.25. Prove that every compact metric space K has a countable base, and that K is therefore separable. *Hint:* For every positive integer n , there are finitely many neighborhoods of radius $1/n$ whose union covers K .⁵⁵

2.26. Let X be a metric space in which every infinite subset has a limit point. Prove that X is compact. *Hint:* By Exercises 2.23 and 2.24, X has a countable base. It follows that every open cover of X has a countable subcover $\{G_n\}$, $n = 1, 2, 3, \dots$. If no finite subcollection of the $\{G_n\}$ covers X , then the complement F_n of $G_1 \cup \dots \cup G_n$ is nonempty for each n , but $\bigcap_{n=1}^{\infty} F_n$ is empty. If E is a set which contains a point from each F_n , consider a limit point of E , and obtain a contradiction.⁵⁶

2.27. Define a point p in a metric space X to be a *condensation point* of a set $E \subset X$ if every neighborhood of p contains uncountably many points of E .

Suppose $E \subset \mathbb{R}^k$, E is uncountable, and let P be the set of all condensation points of E . Prove that P is perfect and that at most countably many points of E are not in P . In other words, show that $P^c \cap E$ is at most countable. *Hint:* Let $\{V_n\}$ be a countable base of $\mathbb{R}^k$, let W be the union of those V_n for which $E \cap V_n$ is at most countable, and show that $P = W^c$.⁵⁷

2.28. Prove that every closed set in a separable metric space is the union of a (possibly empty) perfect set and a set which is at most countable. (*Corollary:* Every countable closed set in $\mathbb{R}^k$ has isolated points.) *Hint:* Use Exercise 2.27.

2.29. Prove that every open set in $\mathbb{R}^1$ is the union of an at most countable collection of disjoint segments. *Hint:* Use Exercise 2.22.⁵⁸

2.30. Imitate the proof of Theorem 2.43 to obtain the following result: If $\mathbb{R}^k = \bigcup_{n=1}^{\infty} F_n$, where each F_n is a closed subset of $\mathbb{R}^k$, then at least one F_n has a nonempty interior.

Equivalent statement: If G_n is a dense open subset of $\mathbb{R}^k$ for $n = 1, 2, 3, \dots$, then $\bigcap_{n=1}^{\infty} G_n$ is not empty (in fact, it is dense in $\mathbb{R}^k$).

(This is a special case of Baire's theorem; see Exercise 3.22 for the general case.)⁵⁹

52 Rational points are countable by 2.13 and dense by 1.20(b) applied coordinatewise. This is the first link of the 2.22–2.26 chain, and it is where Chapter 1's density theorem is finally spent.

53 A countable base is what lets you replace an arbitrary open cover by a countable one — the crucial reduction in 2.26. Countability of the family comes from 2.13 (pairs: centre and radius).

54 The process must stop because an infinite δ -separated set has no limit point — any two of its points stay δ apart, so no neighbourhood of radius $\delta/2$ can hold two of them. This “greedy separation” construction is worth learning; it appears again in proofs of total boundedness.

55 Direct from 2.32: the balls of radius $1/n$ about every point form an open cover, so finitely many suffice. Taking all n gives a countable family.

56 This completes the claim Rudin makes in the text after 2.41: (b) and (c) are equivalent in any metric space. Only (a) is special to $\mathbb{R}^k$. The exercise is genuinely hard and genuinely important; do 2.22–2.25 first, since each is used. Tao proves the same equivalence as his Theorem 1.5.8, taking the sequential property as the definition and deriving the cover property — the reverse of Rudin's order.

57 “Limit point” upgraded from “infinitely many nearby” (2.20) to “uncountably many nearby.” The result says an uncountable set is a perfect set plus countable debris — the Cantor–Bendixson theorem, and the reason 2.28 follows.

58 The structure theorem for open subsets of the line, and it has no analogue in $\mathbb{R}^2$. Take the maximal segment inside the set around each point — these are the connected components, by 2.47 — and each contains a rational, so there are at most countably many.

59 Baire's theorem, and the most consequential exercise in the chapter. It says $\mathbb{R}^k$ cannot be assembled from countably many "thin" closed pieces — a completeness statement, ultimately resting on 2.36 exactly as 2.43 did. It is the engine behind the uniform boundedness principle and the open mapping theorem in functional analysis, and it gives a one-line proof that $\mathbb{R}$ is uncountable: a countable set is a countable union of points, each closed with empty interior.

Appendix to Chapter 2

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 2 is thirty pages of topology with no payoff: every application waits for Chapter 4. He also makes two economies that cost the reader later — he defines *limit point* once and never names the two other formulations he goes on to use, and he proves compactness of a cell by an inlined bisection whose key step is never isolated as a lemma. The Chinese texts organise the same material around *what each theorem buys*, and Tao and Yu Pin carry the metric-space theory past the point where Rudin stops.

What is here.

A. Cluster points. Three inequivalent-looking definitions of "limit point," proved equivalent, and the collision between the set version and the sequence version that Rudin's notation hides.

B. The completeness theorems as tools. Nested intervals with the corollary Rudin omits; Heine–Borel proved through it instead of by raw bisection; Weierstrass twice; and a construction of the least upper bound *from* Cauchy completeness — the one direction Rudin never runs, since he takes LUB as an axiom.

C. The Heine–Borel Rudin declines. At 2.41 he proves the equivalences *in $\mathbb{R}^k$ only*. The general metric-space theorem exists — compact = complete + totally bounded — and it names exactly which property $\mathbb{R}^k$ was supplying.

D. Two notions of smallness. Cardinality is not the only way a set can be small. Baire's theorem supplies the topological notion, and the two disagree in both directions — with Rudin's own Cantor set as one witness.

E. What survives without a metric. Which of Chapter 2 is really about distance, and which is about open sets alone.

How to read it. Section A is worth reading before Chapter 3, because the confusion it settles is the one that bites at 3.5–3.6.

Section B is best read after 2.41, as a second pass over the same theorems by different routes; C and D after the whole chapter. E is optional and answers the question “why a metric at all?” The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — LIMIT POINTS, CLUSTER POINTS, AND A COLLISION

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. Definition 2.18(b) fixes one meaning of *limit point*: every neighbourhood of p meets E in a point other than p . Theorem 2.20 then proves a second — every neighbourhood meets E in *infinitely many* points — and uses it to conclude that a finite set has no limit points.

The gap. There is a third formulation, the one that says p is the limit of a sequence of *distinct* points of E , and Rudin never states it. He nevertheless relies on it constantly from Chapter 3 onward: every argument that begins “choose a sequence in E converging to p ” is using it, and every such argument silently needs the chosen points to be distinct.

Where the gap bites. Twice, and both times painfully. First at 3.5–3.6, where Rudin passes between statements about *sets* (2.42: an infinite subset of a compact set has a limit point) and statements about *sequences* (3.6(b): a bounded sequence in $\mathbb{R}^k$ has a convergent subsequence) as though they were the same theorem. They are not — a sequence has a range, and the range can be finite while the sequence has several subsequential limits. Second at 4.2, where continuity is tested against sequences and one must know which points those sequences may repeat.

What this section supplies. All three formulations, proved equivalent, with the recursion that forces distinctness written out; and the one-line example that separates the set notion from the sequence notion, so that the translation between 2.42 and 3.6(b) can be made carefully instead of tacitly.

S2.1 Theorem. (*Three definitions of a cluster point.*) Let $E \subset \mathbb{R}^1$ and $\xi \in \mathbb{R}^1$. The following are equivalent.

- (a) Every neighbourhood of ξ contains *infinitely many* points of E .
- (b) Every neighbourhood of ξ contains a point of E other than ξ — Rudin’s Definition 2.18(b).

- (c) *There is a sequence (x_n) in E whose terms are **pairwise distinct** with $x_n \rightarrow \xi$.*

SOURCE: ECNU 聚点 Def. 2, 2', 2'' (bk. pp. 151–152). Rudin states (b) and proves (a) as 2.20; (c) he never states.

The one implication with content. That (a) $\Rightarrow$ (b) and (c) $\Rightarrow$ (a) are immediate. For (b) $\Rightarrow$ (c), build the sequence recursively, and *force* distinctness by shrinking the radius past the previous point: having chosen x_{n-1} , apply (b) with

$$\varepsilon_n = \min\left\{\frac{1}{n}, |\xi - x_{n-1}|\right\}$$

to obtain $x_n \in E$ with $0 < |x_n - \xi| < \varepsilon_n$. Then $x_n \rightarrow \xi$, and no point can repeat, since each is strictly nearer ξ than its predecessor. $\square$

WHY THE RECURSION NEEDS THAT PARTICULAR RADIUS

The obvious recursion — take $|x_n - \xi| < 1/n$ — gives convergence but not distinctness: nothing stops the same point being chosen twice. The extra clause $|\xi - x_{n-1}|$ in the minimum makes the distances strictly decreasing, and a strictly decreasing sequence of distances cannot repeat a point.

This is exactly the step a reader of Rudin has to invent unaided. Theorem 2.20 gives him “infinitely many points in every neighbourhood”; every later argument that says “choose a sequence in E converging to ξ ” silently needs those points to be distinct, and that is what the recursion supplies. Rudin uses the fact at 2.20’s corollary (a finite set has no limit points), at 3.6, and throughout Chapter 4.

S2.2 Definition. (*Cluster point of a sequence.*) a is a cluster point of the sequence (x_n) if every neighbourhood of a contains x_n for **infinitely many indices** n — terms counted with repetition. Equivalently, a is the limit of some subsequence.

SOURCE: ECNU §7.2 Def. 1 and the following 注 (bk. p. 156).

THE SET OF TERMS IS NOT THE SEQUENCE

The two notions disagree, and the cleanest witness is one line. Take

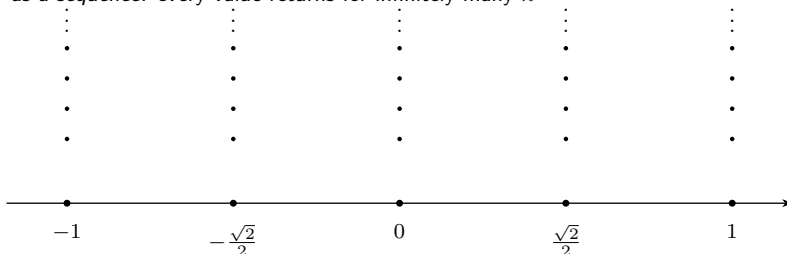
$$x_n = \sin \frac{n\pi}{4}.$$

As a *sequence* it has five cluster points: $-1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}, 1$, each hit infinitely often. As a *set*, its range is the five-element set $\{-1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}, 1\}$, and a finite set has **no** limit points at all (Rudin’s corollary to 2.20).

So “limit point” in Chapter 2 and “subsequential limit” in 3.5–3.6 are different notions that can disagree completely. Rudin uses both words without ever flagging the difference, and 3.6(b) — every bounded sequence in $\mathbb{R}^k$ has a convergent subsequence — is *not* literally 2.42 applied to the range, because the range may be finite. The bridge is:

a subsequential limit of (x_n) is either a limit point of the range, or a value taken infinitely often. Rudin's proof of 3.6(b) quietly splits into exactly those two cases.

as a sequence: every value returns for infinitely many n



as a set: just these five points — no limit point at all

$x_n = \sin(n\pi/4)$ has five cluster points as a sequence and none as a set. Multiplicity is exactly the information the range throws away — which is why Rudin's 2.42 (about sets) does not directly give 3.6(b) (about sequences).

NOT IN RUDIN

SUPPLEMENT B — THE COMPLETENESS THEOREMS AS TOOLS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. He assumes the least-upper-bound property as an axiom (1.19) and derives everything from it in one direction: 1.20 (Archimedes and density), 2.38 (nested compact sets), 2.40 (k -cells are compact), 2.41 (Heine–Borel in $\mathbb{R}^k$), 3.11 (completeness).

The gap. A one-directional development hides the fact that these statements are *equivalent*. Chapter 1's comparison box lists six formulations of completeness and notes that different books axiomatise different ones; but a list of equivalent statements teaches nothing until you have seen the implications run. Rudin runs them only outward from LUB, so a reader has no way to discover that LUB can be recovered from Cauchy completeness — or what extra hypothesis that recovery needs.

Where the gap bites. At 2.40 the crucial step — that a nested family with vanishing diameters is eventually swallowed by any neighbourhood of its common point — is derived inline, in the middle of a k -dimensional argument, and never named. It is needed again at 2.42, at 3.10, and in every later bisection argument, and each time it is rebuilt from scratch.

What this section supplies. That step, isolated as a lemma with its corollary (S2.3), after which Heine–Borel is three lines

(S2.4) and Weierstrass is four (S2.5); a proof of Bolzano–Weierstrass that uses no bisection at all and secretly computes $\liminf$ (S2.6); and the missing implication, a construction of $\sup$ from Cauchy completeness plus Archimedes (S2.7), which shows precisely why the Archimedean hypothesis cannot be dropped.

S2.3 Theorem. (*Nested intervals, with its corollary.*) Let $[a_{n+1}, b_{n+1}] \subset [a_n, b_n]$ for all n and $b_n - a_n \rightarrow 0$. Then there is exactly one ξ with $\xi \in [a_n, b_n]$ for every n ; and moreover, for every $\varepsilon > 0$ there is an N with

$$[a_n, b_n] \subset (\xi - \varepsilon, \xi + \varepsilon) \quad (n > N).$$

SOURCE: ECNU 区间套定理 7.1 and its 推论 (bk. pp. 150–151).

Proof. The a_n increase and every b_m is an upper bound for them: if $m \leq n$ then $a_n \leq b_n \leq b_m$, and if $m > n$ then $a_n \leq a_m \leq b_m$. So $\{a_n\}$ is nonempty and bounded above, and by 1.19 it has a supremum ξ , with $a_n \leq \xi \leq b_n$ for every n . If ξ' also lay in every $[a_n, b_n]$ then $|\xi - \xi'| \leq b_n - a_n \rightarrow 0$, so $\xi' = \xi$.

For the corollary, given $\varepsilon > 0$ choose N with $b_N - a_N < \varepsilon$. For $n > N$ the interval $[a_n, b_n]$ is contained in $[a_N, b_N]$, has length less than ε , and contains ξ ; hence every one of its points lies within ε of ξ . $\square$

Rudin's 2.38 gives nonempty intersection for nested compact sets and stops there. The corollary — eventually the whole interval, not merely the point, sits inside the neighbourhood — is what does the work in every application below.⁶⁰

CLOSED IS NOT DECORATION

The intervals must be closed. The open intervals $(0, 1/n)$ are nested with lengths tending to 0 and have **empty** intersection. Boundedness matters too: the closed sets $[n, \infty)$ are nested with empty intersection. Rudin's 2.38 requires compactness, which packages both hypotheses at once; seeing the two failures separately is what makes the packaging legible.

S2.4 Theorem. (*Heine–Borel, via nested intervals.*) Every open cover of $[a, b]$ has a finite subcover.

SOURCE: ECNU 有限覆盖定理 7.3 (bk. p. 153).

HOW THIS PROOF DIFFERS FROM RUDIN'S 2.40

1. **Same start.** Suppose no finite subcover exists. Bisect $[a, b]$; at least one half again admits no finite subcover. Iterate, producing nested intervals with lengths $\rightarrow 0$.
2. **The step that is now free.** Let ξ be the common point.

⁶⁰ The proof of the corollary is one line: $\xi \in [a_n, b_n]$ and $b_n - a_n \rightarrow 0$, so both endpoints are within ε of ξ once $b_n - a_n < \varepsilon$. Trivial to prove, easy to forget you need, and Rudin never names it — so at 2.40 he re-derives it inline.

It lies in some cover element (α, β) . By the *corollary* to S2.3, $[a_n, b_n] \subset (\alpha, \beta)$ for large n — so that interval is covered by a *single* member, contradicting the choice of intervals with no finite subcover.

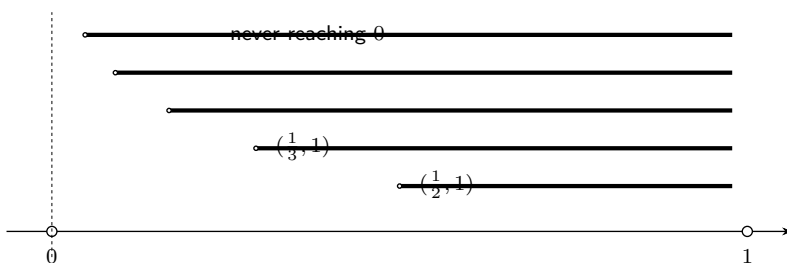
3. **What changed.** Rudin's 2.40 proves the same thing but derives step 2 inline, in the middle of a k -dimensional argument, using $\text{diam } Q_n \rightarrow 0$. Naming the lemma first turns the proof into three lines and makes visible that bisection is used only to *manufacture* a nested family — any other method of manufacturing one would do.
4. **Audit.** Closedness of $[a, b]$ enters once, in S2.3; boundedness enters once, to start the bisection. Both hypotheses are spent exactly where the pitfall above says they must be.

WHERE THE THEOREM FAILS, MADE CONCRETE

On $(0, 1)$ the cover

$$\left\{ \left(\frac{1}{n+1}, 1 \right) : n = 1, 2, 3, \dots \right\}$$

works and no finite subfamily does: the members increase, so a finite subfamily is its largest member, which omits everything below $\frac{1}{N+1}$. Rudin gives the analogous fact at 2.32's examples; the point worth keeping is the mechanism — a totally ordered cover with no largest element can only be finitely reduced if one member already suffices.



The cover of $(0, 1)$ that admits no finite subcover. The members are nested upward, so any finite subfamily equals its largest member and misses a whole interval at the left end. Compactness fails exactly at the point the domain is missing.

S2.5 Theorem. (Weierstrass.) Every bounded infinite subset of $\mathbb{R}^1$ has a cluster point.

SOURCE: ECNU 聚点定理 7.2, with two proofs (bk. pp. 152–153).

Proof. Let E be infinite with $E \subset [-M, M]$. Bisect: at least one of the two halves contains infinitely many points of E , since otherwise E would be finite. Call it $[a_1, b_1]$ and repeat, obtaining nested intervals $[a_n, b_n]$, each containing infinitely many points of E , with $b_n - a_n = 2M/2^n \rightarrow 0$.

By S2.3 there is a point ξ common to all of them. Given $\varepsilon > 0$, the

corollary to S2.3 puts $[a_n, b_n]$ inside $(\xi - \varepsilon, \xi + \varepsilon)$ for all large n ; and $[a_n, b_n]$ contains infinitely many points of E . So every neighbourhood of ξ contains infinitely many points of E , which is S2.1(a). $\square$

Two proofs, landing on two different definitions

First proof (bisection). $E \subset [-M, M]$; bisect, always keeping a half containing infinitely many points of E . By S2.3 and its corollary, every neighbourhood of the common point ξ contains some $[a_n, b_n]$, hence infinitely many points of E . This lands on definition (a) of S2.1.

Second proof (via sequences). Choose pairwise distinct $x_n \in E$; the sequence is bounded, so some subsequence converges, say to x_0 ; then every neighbourhood of x_0 contains infinitely many of the x_n . This lands on definition (c).

The two proofs are the same argument in different costumes, and S2.1 is what lets you move between them. ECNU then states the dictionary Rudin leaves implicit: *the sequence form of Bolzano–Weierstrass is the set form applied to the set of terms, with “infinitely many points” read as “infinitely many indices.”* Rudin’s 2.42 and 3.6(b) sit twenty pages apart with no remark connecting them — and, as the pitfall at S2.2 shows, the translation needs the care that remark would have supplied.

S2.6 Theorem. (*Bolzano–Weierstrass without bisection.*) Every bounded sequence (a_n) of reals has a convergent subsequence.

SOURCE: USTC 定理 8 (bk. p. 12).

The monotone-tail proof. Put $\alpha_n = \inf\{a_n, a_{n+1}, \dots\}$. The sequence (α_n) is increasing and bounded above, hence converges; call the limit α . Now build indices: by the definition of infimum there is n_1 with $\alpha_1 \leq a_{n_1} < \alpha_1 + 1$; having chosen n_k , there is $n_{k+1} > n_k$ with

$$\alpha_{n_k} \leq a_{n_{k+1}} < \alpha_{n_k} + \frac{1}{k+1}.$$

Since $\alpha_{n_k} \rightarrow \alpha$, the squeeze gives $a_{n_k} \rightarrow \alpha$. $\square$

What this proof reveals that Rudin’s route hides

Rudin reaches Bolzano–Weierstrass only through the cell machinery: 2.40 (k -cells are compact) $\rightarrow$ 2.42 (infinite subsets of compact sets have limit points) $\rightarrow$ 3.6(b). That path makes the theorem look as though it *needs* compactness, and in $\mathbb{R}^k$ it is the efficient route. On the line it needs nothing but the monotone convergence theorem. And the auxiliary sequence is not arbitrary: $\alpha_n = \inf_{k \geq n} a_k$ is precisely the quantity whose limit Rudin will call $\liminf a_n$ at 3.16. So this proof silently establishes something stronger than it claims — *the subsequential limit it produces is the lower limit* — which is Rudin’s 3.17(a) arriving a chapter early and for free.

Keep the contrast: bisection halves the *domain*, the monotone-tail argument shaves the *index set*. Bisection generalises to $\mathbb{R}^k$; the tail argument does not, because it needs the order.

S2.7 Theorem. (*The least upper bound, constructed from Cauchy completeness.*) Assume $\mathbb{R}$ is Archimedean and that every Cauchy sequence of reals converges. Then every nonempty set bounded above has a supremum.

SOURCE: ECNU 例 3 (bk. pp. 154–155), the $6 \Rightarrow 1$ step of the completeness cycle.

THE CONSTRUCTION

1. **Manufacture near-misses.** Let $S \neq \emptyset$ be bounded above. For each n , the Archimedean property supplies an integer k_n such that $\lambda_n = k_n/n$ is an upper bound of S while $\lambda_n - 1/n$ is not. (Walk the multiples of $1/n$ upward until they first clear S .)
2. **Show the near-misses are Cauchy.** Since λ_m is an upper bound and $\lambda_n - 1/n$ is not, some $a' \in S$ has $\lambda_m \geq a' > \lambda_n - 1/n$, so $\lambda_n - \lambda_m < 1/n$. By symmetry $|\lambda_m - \lambda_n| < \max\{1/m, 1/n\}$.
3. **Cash in completeness.** So $\lambda_n \rightarrow \lambda$ for some real λ .
4. **Identify the limit.** λ is an upper bound, being a limit of upper bounds. And it is least: given $\delta > 0$ choose n with $1/n < \delta/2$ and $\lambda_n > \lambda - \delta/2$; since $\lambda_n - 1/n$ is not an upper bound, some $a' \in S$ exceeds it, hence exceeds $\lambda - \delta$.
5. **Audit.** The Archimedean property is spent once, in step 1; completeness once, in step 3. Neither is decorative — see the pitfall.

Why Rudin never does this, and why it is worth seeing

Rudin takes the LUB property as an *axiom* (1.19) and derives the Archimedean property from it (1.20a). He therefore never shows the converse direction, and a reader of Rudin alone could reasonably conclude that “complete” has only one meaning.

It has two, and they are not the same. *Cauchy completeness alone does not imply the LUB property*: there are Cauchy-complete ordered fields without it — non-Archimedean ones. That is why step 1 above needs the Archimedean hypothesis, and it is exactly the gap Yu Pin’s axiom list (field + order + Archimedean + nested intervals) is built to expose. Chapter 1’s comparison box lists the equivalences; this is the proof of the one link that Rudin’s development makes structurally invisible.

NOT IN RUDIN

HEINE–BOREL RUDIN DECLINES

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. Theorem 2.41 proves that for a set in $\mathbb{R}^k$, three conditions coincide: compact, closed and bounded, and every infinite subset has a limit point. He states the restriction to $\mathbb{R}^k$ emphatically, and Exercise 2.16 supplies a metric space where closed and bounded fails to give compact.

The gap. The restriction is a warning, not an explanation. A reader finishes 2.41 knowing that “closed and bounded” is the wrong condition in general, but not what the right one is — and therefore not what property $\mathbb{R}^k$ was quietly contributing.

Where the gap bites. Everywhere Rudin later works in a space that is not $\mathbb{R}^k$. At 7.23 (Arzelà–Ascoli) he must characterise the compact subsets of a function space, and the characterisation is exactly the general theorem below — pointwise boundedness and equicontinuity are total boundedness in disguise. Chapter 11’s $\mathcal{L}^2$ is another such space. Without the general statement, each of these looks like a new trick.

What this section supplies. The general theorem — compact = complete + totally bounded — with proof, so that 2.41 decodes: *closed* was supplying completeness and *bounded* was supplying total boundedness, and only the second implication is special to $\mathbb{R}^k$. Also the equivalence of the two compactnesses in a metric space, which Rudin proves only inside $\mathbb{R}^k$.

S2.8 Definition. A metric space X is **complete** if every Cauchy sequence in X converges in X ; it is **totally bounded** if for every $\varepsilon > 0$ finitely many balls of radius ε cover X .

SOURCE: Yu Pin 定义 35 (p.76); Tao II, Ex. 1.5.10 (bk. p.26). Rudin defines completeness at 3.11 and total boundedness nowhere.

S2.9 Theorem. (Heine–Borel for metric spaces.) A metric space is compact if and only if it is complete and totally bounded.

SOURCE: Tao II, Ex. 1.5.10(b),(c).

Proof. Throughout, use S2.10 to read “compact” as “every sequence has a convergent subsequence.”

Compact $\Rightarrow$ totally bounded. Given $\varepsilon > 0$, the balls $B(x, \varepsilon)$, $x \in X$, form an open cover; a finite subcover is a finite family of ε -balls covering X .

Compact $\Rightarrow$ complete. Let (x_n) be Cauchy. Some subsequence converges, say $x_{n_k} \rightarrow x$. Given $\varepsilon > 0$ choose N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$, and then k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$; for $n \geq N$ the triangle inequality gives $d(x_n, x) < \varepsilon$. So the whole sequence converges.

Complete and totally bounded $\Rightarrow$ compact. Let (x_n) be any sequence. Cover X by finitely many balls of radius 1; one of them contains x_n for infinitely many n , giving a subsequence S_1 lying in a set of diameter at most 2. Cover X by finitely many balls of radius $\frac{1}{2}$; one contains infinitely many terms of S_1 , giving a subsequence $S_2 \subset S_1$ of diameter at most 1. Continuing, we obtain $S_1 \supset S_2 \supset \cdots$, with S_j of diameter at most 2^{2-j} .

Let y_k be the k th term of S_k . For $i, j \geq k$ both y_i and y_j belong to S_k , so $d(y_i, y_j) \leq 2^{2-k}$: the diagonal sequence (y_k) is Cauchy, and by completeness it converges. It is a subsequence of (x_n) , so X is sequentially compact. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** Two properties must be produced from one, and then one from two. The forward direction is cheap because compactness is a hypothesis about *every* cover and about *every* sequence, so one simply applies it to a well-chosen instance. The converse is where the work is: nothing is compact yet, so a convergent subsequence has to be manufactured by hand.
2. **Spend the cover hypothesis (forward, part one).** Apply compactness to the cover by all ε -balls. There is nothing to choose — the cover is handed to you by the definition of total boundedness itself.
3. **Spend the sequence hypothesis (forward, part two).** A Cauchy sequence needs only *one* convergent subsequence to converge itself; that upgrade is the same three-line estimate as in 3.11 and is the whole content of “compact implies complete.”
4. **Manufacture nested pigeonholes (converse).** Total boundedness gives finitely many balls at each scale, and finitely many boxes holding infinitely many terms means one box holds infinitely many. Halving the radius each time makes the boxes shrink geometrically. This is pattern 3 of Chapter 1’s strategy index — when the hypothesis is finiteness, pigeonhole.
5. **Take the diagonal.** No single S_j converges, so no one of them is the answer; the diagonal belongs to every S_k from the k th place on and is therefore Cauchy. The diagonal is what turns a *nested family* of subsequences into a *single* subsequence.
6. **Cash in completeness.** Cauchy becomes convergent, and only here. Rudin performs the identical manoeuvre at 7.23 to extract a convergent subsequence of functions.
7. **Audit.** Total boundedness is spent once, in step 4, to get the

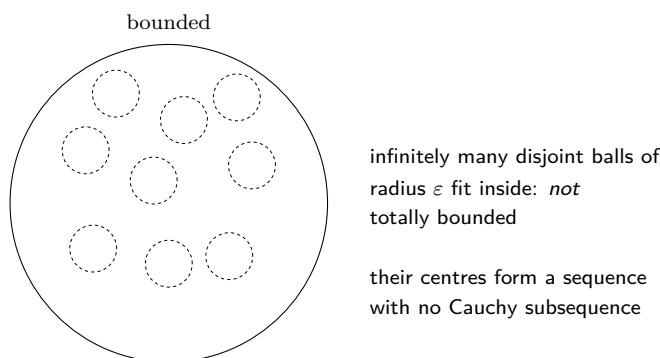
boxes; completeness once, in step 6, to convert the diagonal's Cauchy property into a limit. Neither is decorative: the sequence $(1/n)$ in $(0, 1]$ is totally bounded and not complete, and the unit vectors of the picture below are complete and not totally bounded, and neither space is compact.

Reading Rudin's 2.41 through this

Now Rudin's restriction to $\mathbb{R}^k$ decodes. "Closed and bounded" is doing two separate jobs there:

Closed in $\mathbb{R}^k$ gives **completeness** — a closed subset of a complete space is complete, and $\mathbb{R}^k$ is complete by 3.11. *Bounded* in $\mathbb{R}^k$ gives **total boundedness**, because a bounded set fits in a cell and a cell is chopped into finitely many pieces of any prescribed size.

The second implication is the one that fails elsewhere, and it fails for a reason with a name: in infinite dimensions, bounded does not imply totally bounded. Rudin's Exercise 2.16 — the set $\{p \in \mathbb{Q} : 2 < p^2 < 3\}$, closed and bounded in $\mathbb{Q}$ but not compact — diagnoses the *other* failure, incompleteness. Both diseases exist, and 2.41 is the statement that $\mathbb{R}^k$ has neither.



Bounded but not totally bounded — the failure Rudin's 2.41 rules out by working in $\mathbb{R}^k$. In ℓ^1 the unit vectors $e^{(n)}$ are at mutual distance 2, so the balls of radius 1 about them are disjoint; the set is closed, bounded, and lives in a complete space, yet no subsequence converges. Total boundedness is precisely the hypothesis that forbids this picture.

S2.10 Theorem. (The two compactnesses agree.) For a metric space X : every open cover of X has a finite subcover if and only if every sequence in X has a convergent subsequence.

SOURCE: Yu Pin, p. 114 (both directions); Tao II, Thm 1.5.8 and Ex. 1.5.11 — Tao defines compactness sequentially, the mirror image of Rudin.

Proof. *Covers $\Rightarrow$ sequences.* Suppose (x_k) had no convergent subsequence. Then no point x is a subsequential limit, so each x has a radius $r_x > 0$ for which $B(x, r_x)$ contains x_k for only finitely many indices k . These balls cover X ; a finite subcover would then account for only finitely many indices altogether, which is absurd.

Sequences $\Rightarrow$ covers. Let $\{U_\alpha\}$ be an open cover. First, X is totally bounded: if some $\varepsilon > 0$ admitted no finite cover by ε -balls, one could choose x_1 , then x_2 outside $B(x_1, \varepsilon)$, then x_3 outside the first two balls, and so on, producing a sequence with mutual distances at least ε and hence no convergent subsequence.

Second, the cover has a Lebesgue number (S4.19 in the Chapter 4 appendix): if not, then for each n some $B(x_n, 1/n)$ lies in no U_α ; passing to a convergent subsequence $x_{n_k} \rightarrow x$ and choosing β and r with $B(x, r) \subset U_\beta$, we get $B(x_{n_k}, 1/n_k) \subset B(x, r)$ for large k — a contradiction.

Now cover X by finitely many balls of radius δ , a Lebesgue number. Each lies inside a single U_α , and those finitely many U_α cover X . $\square$

Rudin proves one direction inside $\mathbb{R}^k$ (2.41, via 2.37) and never states the general equivalence.⁶¹

61 *The forward direction is slick: if (x_k) had no convergent subsequence, every tail $\{x_k : k \geq m\}$ would be closed, so the complements $U_m = X \setminus \{x_k : k \geq m\}$ form an open cover with no finite subcover. The reverse direction is the one that needs a uniform scale — the Lebesgue number lemma, stated as S4.19 in the Chapter 4 appendix, where Rudin's 4.19 needs it too.*

NOT IN RUDIN

SUPPLEMENT D — TWO NOTIONS OF SMALLNESS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. The chapter opens with cardinality (2.1–2.14) and closes with the Cantor set (2.44), an uncountable set that is, in his words, of measure zero. Size is measured by counting.

The gap. Counting is not the only measure of size, and Rudin's own examples show it is not the useful one for topology. $\mathbb{Q}$ is countable — small by counting — yet meets every interval. The Cantor set is uncountable — large by counting — yet contains no interval at all. The notion that separates them is nowhere in Chapter 2, though every ingredient for it is: closure, interior, density.

Where the gap bites. At Exercise 3.22, where Rudin poses Baire's theorem with no indication of what it is for; at 7.18, where the existence of a nowhere-differentiable continuous function looks like a lucky construction rather than the typical case; and in the Chapter 4 appendix, where the question “can a function be continuous exactly on $\mathbb{Q}$?” cannot even be posed without it.

What this section supplies. Cantor's theorem in general form and Schröder–Bernstein, which make cardinality an order relation rather than a collection of tricks; then Baire's theorem with proof, and the table showing that the two notions of smallness disagree in both directions.

S2.11 Theorem. (Cantor.) *For every set X , there is no surjection $X \rightarrow 2^X$; in particular X and its power set never have the same cardinality.*

SOURCE: Tao I, Thm 8.3.1 (bk. p. 195). Rudin's 2.14 is the case $X = \mathbb{N}$.

Proof. Let $f : X \rightarrow 2^X$ be any map and put

$$A = \{x \in X : x \notin f(x)\}.$$

If $A = f(x_0)$ for some x_0 , then $x_0 \in A$ means $x_0 \notin f(x_0) = A$, while $x_0 \notin A$ means $x_0 \in f(x_0) = A$. Both are impossible, so A is not in the range of f . $\square$

The proof is Rudin's 2.14 with the decimal expansions stripped away: given $f : X \rightarrow 2^X$, the set $A = \{x : x \notin f(x)\}$ is not $f(x)$ for any x .⁶²

S2.12 Theorem. (Schröder–Bernstein.) *If there are injections $A \rightarrow B$ and $B \rightarrow A$, then there is a bijection $A \rightarrow B$.*

SOURCE: Tao I, Ex. 8.3.3; Yu Pin p. 38 gives a different proof by least fixed point.

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. Put

$$C_0 = A \setminus g(B), \quad C_{n+1} = g(f(C_n)), \quad C = \bigcup_{n \geq 0} C_n,$$

and define

$$h(a) = \begin{cases} f(a) & (a \in C), \\ g^{-1}(a) & (a \notin C). \end{cases}$$

The second line makes sense: $a \notin C$ gives $a \notin C_0$, so $a \in g(B)$, and g is injective, so $g^{-1}(a)$ is a single point.

h is injective. Both f and g^{-1} are, so the only risk is $f(a) = g^{-1}(a')$ with $a \in C$, $a' \notin C$. But then $a \in C_n$ for some n and $a' = g(f(a)) \in C_{n+1} \subset C$, a contradiction.

h is onto. Let $b \in B$. If $g(b) \in C$, then $g(b) \in C_{n+1}$ for some n — it cannot lie in C_0 — so $g(b) = g(f(c))$ for some $c \in C_n$, whence $b = f(c) = h(c)$ by injectivity of g . If $g(b) \notin C$, then $h(g(b)) = g^{-1}(g(b)) = b$. $\square$

This is what makes cardinality an order relation rather than a collection of tricks: it is antisymmetry.⁶³

S2.13 Theorem. (Baire.) *In a complete metric space, the intersection of countably many dense open sets is dense. Equivalently, a countable union of closed sets with empty interior has empty interior.*

SOURCE: Yu Pin 定理 160–161 (pp. 296–297). Rudin gives it as Exercise 3.22.

Proof. Let $U_1, U_2, \dots$ be dense open sets in the complete space X , put $V = \bigcap_n U_n$, and let W be any nonempty open set; we must produce a point of $V \cap W$.

Since U_1 is dense and open, $U_1 \cap W$ is nonempty and open, so it contains a closed ball $\overline{B}(x_1, r_1)$ with $0 < r_1 < 1$. Having chosen $\overline{B}(x_n, r_n)$, the set $U_{n+1} \cap B(x_n, r_n)$ is nonempty and open, so it contains a closed ball $\overline{B}(x_{n+1}, r_{n+1})$ with $0 < r_{n+1} < r_n/2$.

⁶² Rudin proves $2^{\mathbb{N}}$ is uncountable by a diagonal argument on sequences of 0s and 1s, which is this proof in a costume. Stated for general X it yields an infinite ladder of cardinalities, and with it the question Rudin never raises — whether anything sits strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$. That is the continuum hypothesis, and Gödel and Cohen showed it is undecidable from the usual axioms.

⁶³ Rudin compares cardinalities throughout Chapter 2 — “at most countable,” “ E is equivalent to $\mathbb{N}$ ” — and never needs to state that $|A| \leq |B|$ and $|B| \leq |A|$ force $|A| = |B|$, because his cases are concrete enough to exhibit bijections directly. The general fact is not obvious and the proof is genuinely clever.

The balls are nested, so for $m > n$ we have $x_m \in \overline{B}(x_n, r_n)$ and hence $d(x_m, x_n) \leq r_n < 2^{-n+1}$: the centres form a Cauchy sequence. By completeness $x_n \rightarrow x$. For each n , all x_m with $m \geq n$ lie in the closed set $\overline{B}(x_n, r_n)$, so x does too; therefore $x \in U_n$ for every n , and $x \in \overline{B}(x_1, r_1) \subset W$. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** “ V is dense” means V meets every nonempty open W , so the goal is to *produce a point* lying in infinitely many sets at once. The only machinery that produces a point from infinitely many conditions is a nested construction followed by completeness — the same shape as S2.3 and 2.38.
2. **Use density to move, openness to get room.** At each stage density guarantees U_{n+1} reaches into the ball we are standing in; openness guarantees the intersection has room for a whole closed ball. Both hypotheses are used at every step, and neither can be dropped: a dense set that is not open, like $\mathbb{Q}$, leaves no room.
3. **Halve the radii.** Forcing $r_{n+1} < r_n/2$ makes the centres Cauchy. Any summable rate would do; halving is the cheapest.
4. **Take *closed* balls.** This is the one design choice with content. The limit x need not lie in any open ball $B(x_n, r_n)$, but it must lie in the closed one, because closed sets contain their limits (2.27). Closing the balls is what makes the intersection catch the limit point.
5. **Cash in completeness.** The centres converge, and the limit lies in every $\overline{B}(x_n, r_n) \subset U_n$, hence in V ; and in $\overline{B}(x_1, r_1) \subset W$.
6. **Audit.** Completeness is spent once, at step 5; density and openness once per stage, at step 2; closedness of the balls once, at step 4. Drop completeness and the theorem is false — in $\mathbb{Q}$, the sets $\mathbb{Q} \setminus \{q_n\}$ are dense and open with empty intersection.

The second notion of smallness, and how it cuts across the first

Call a set *nowhere dense* if its closure has empty interior, and *meagre* if it is a countable union of nowhere dense sets. Baire’s theorem says a complete metric space is not meagre in itself — meagre sets are topologically negligible.

The two notions of smallness are independent, and Chapter 2 already contains both witnesses:

set	cardinality	topologically
$\mathbb{Q} \subset \mathbb{R}$	countable — small	dense, not meagre in itself
the Cantor set (2.44)	uncountable — large	nowhere dense — small

So the Cantor set is large by counting and negligible by topology, while $\mathbb{Q}$ is the reverse. Rudin builds the Cantor set precisely to exhibit an uncountable set of “measure zero” (his phrase at 2.44), which is a third notion again — the one Chapter 11 will develop. Yu Pin’s applications show what the theorem is for: no continuous function on $[a, b]$ can be differentiable with unbounded derivative on every subinterval; the indicator of $\mathbb{Q}$ is not a pointwise limit of continuous functions; and “most” continuous functions are nowhere differentiable, the qualitative form of Rudin’s 7.18. The Chapter 4 appendix uses Baire again, at S4.5, to show no function is continuous exactly on the rationals.

NOT IN RUDIN

SUPPLEMENT E — WHAT SURVIVES WITHOUT A METRIC

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. He defines everything in terms of the distance d : balls, limit points, open sets, compactness, connectedness. Theorem 2.24 then proves that arbitrary unions and finite intersections of open sets are open.

The gap. 2.24 is the moment the metric stops mattering, and Rudin does not say so. From there to the end of the chapter — and through 4.8, 4.14 and 4.22 in the next — the proofs use only the conclusions of 2.24, never the distance that produced them. A reader has no way to tell which results are really about distance and which are about open sets alone.

Where the gap bites. At 2.34 (compact implies closed), which needs a separation hypothesis that $\mathbb{R}^k$ supplies for free; at the equivalence of the two compactnesses (S2.10), which is genuinely metric and fails without it; and at every point where a reader wonders whether uniform continuity could be defined the same way — it cannot, and knowing why saves a false generalisation.

What this section supplies. The definition that takes 2.24 as an axiom, and a table sorting Chapter 2 into what survives and what dies, with the two counterexamples that mark the boundary.

S2.14 Definition. A *topology* on X is a family $\mathcal{F}$ of subsets containing $\emptyset$ and X and closed under arbitrary unions and finite intersections; its members are the open sets. X is **Hausdorff** if distinct points have disjoint neighbourhoods.

SOURCE: Tao II, §2.5 (bk. pp. 39–43); Yu Pin 定义 78 (p. 130).

Which of Rudin's Chapter 2 survives, and which dies

Rudin's 2.24 — arbitrary unions and finite intersections of open sets are open — is not a theorem in this setting but the *definition*. Taking it as the definition, one can ask which of the rest survives.

Survives with no metric

Needs the metric

closure, interior, boundary; the relative topology (2.30); continuity as preimages of open sets (4.8); compactness by finite subcovers; connectedness

Cauchy sequences and completeness; boundedness; uniform continuity; “compact $\Rightarrow$ sequentially compact”

Two of Rudin's results fail outright. *Compact implies closed* (2.34) needs Hausdorff: in the trivial topology $\{\emptyset, X\}$ every subset is compact and only two are closed. And *compact implies sequentially compact* fails even in a Hausdorff space — Tao's Exercise 2.5.8 constructs a compact ordered space containing a sequence with no convergent subsequence, which is exactly why S2.10 above had to say “metric space.”

Uniqueness of limits is the cleanest illustration of what Hausdorff buys: without it a sequence can converge to two different points at once. Rudin gets uniqueness free at 3.2(b) because a metric space is automatically Hausdorff — take balls of radius $d(p, q)/2$.

NOT IN RUDIN

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S THIRTY

Rudin's own Exercises 2.1–2.30 end the chapter proper. They drill his vocabulary: closures, condensation points, separability, the distance function. These twenty-one come from the four comparison texts and do something his cannot — they make the machinery *pay*.

Group I (1–10): the completeness theorems, proved with each other. Rudin derives everything from the least-upper-bound axiom in one direction. These run the implications the other way and, in exercise 1, show what fails over $\mathbb{Q}$ — the exercise that turns the equivalence list of Chapter

64 The exercise that gives the equivalence list of Chapter 1 its point. One example does all four: take rationals increasing to $\sqrt{2}$ — say the decimal truncations. The set is bounded above with no rational supremum, the sequence is monotone bounded and does not converge in $\mathbb{Q}$, it is Cauchy without a limit, and its only candidate cluster point is irrational. That a single witness kills all four is the equivalence, seen from below.

65 Three questions probing exactly where finiteness breaks. Note that (c) succeeds while (b) fails, which isolates the mechanism: the trouble is never “the set is infinite” but “the cover degenerates at a point the set does not contain.” Rudin’s cover examples at 2.32 are one-liners with no such sub-questions.

66 The contrapositive direction Rudin never runs. If bounded infinite E had no cluster point, every $x \in [a, b]$ would have a neighbourhood meeting E in finitely many points; these cover $[a, b]$, so finitely many do, making E finite. Compactness converting “locally finite” into “globally finite” is the whole idea, and it recurs at 4.15.

67 A Cauchy sequence is bounded, so it has a cluster point; then show the whole sequence converges to it, using the Cauchy condition to transfer nearness from the subsequence to the tail. Rudin does exactly this at 3.11, without naming the route. Together with exercise 3 this closes the loop of implications.

68 The “creeping along” proof, and a genuinely different mechanism from S2.4. Let E be the set of $x \in [a, b]$ such that $[a, x]$ admits a finite subcover, and put $c = \sup E$. Show $c \in E$ — some cover element contains c , and it also contains points slightly left of c — and then that $c = b$, since otherwise the same element lets you creep past c . Keep the pattern: to prove something holds on all of $[a, b]$, take the supremum of where it holds and show that supremum cannot be interior. It reappears in the proof of the mean value theorem and in Chapter 5’s appendix.

1 into a statement about $\mathbb{R}$ specifically.

Group II (11–16): metric spaces past Rudin’s stopping point. Completion, total boundedness, the contraction mapping principle, and the closed bounded complete set that is still not compact.

Group III (17–20): size. Uncountability without a diagonal argument, Cantor’s theorem, Schröder–Bernstein, and a compact space containing a sequence with no convergent subsequence.

Group IV (21): topology as a tool. Furstenberg’s proof that there are infinitely many primes, using nothing but Rudin’s 2.23.

Marked ★: harder, or requiring a result from a later chapter.

S2.1. Give examples showing that over the rationals, each of the following fails: the least-upper-bound principle; the monotone convergence theorem; the cluster-point (Bolzano–Weierstrass) theorem; and the Cauchy criterion.⁶⁴

SOURCE: ECNU 习题 7.1 #4 (bk. p. 155).

S2.2. Let $H = \{(\frac{1}{n+2}, \frac{1}{n}) : n = 1, 2, 3, \dots\}$. (a) Does H cover $(0, 1)$? (b) Can finitely many members of H cover $(0, \frac{1}{2})$? (c) Can finitely many cover $(\frac{1}{100}, 1)$?⁶⁵

SOURCE: ECNU 习题 7.1 #5 (bk. p. 155).

S2.3. Prove the cluster-point theorem (S2.5) using the finite covering theorem (S2.4).⁶⁶

SOURCE: ECNU 习题 7.1 #8 (bk. p. 155).

S2.4. Prove the Cauchy criterion using the cluster-point theorem.⁶⁷

SOURCE: ECNU 习题 7.1 #9 (bk. p. 155).

S2.5. ★ Prove the finite covering theorem using the least-upper-bound principle, without bisection.⁶⁸

SOURCE: ECNU 总练习题 7 #2 (bk. p. 160).

S2.6. Prove that a continuous real function on $[a, b]$ is uniformly continuous, using the finite covering theorem.⁶⁹

SOURCE: ECNU 习题 7.1 #11 (bk. p. 155). Companion #10: prove the root-existence theorem by the same method.

S2.7. Let (x_n) be monotone. Prove that if it has a cluster point, that cluster point is unique and equals $\sup_n x_n$ (for increasing (x_n)).⁷⁰

SOURCE: ECNU 习题 7.1 #7 (bk. p. 155).

S2.8. Prove that the set of cluster points of the closed interval $[a, b]$ is $[a, b]$ itself.⁷¹

SOURCE: ECNU 习题 7.1 #6 (bk. p. 155).

S2.9. Sharpen the nested interval theorem: if $a_n < a_{n+1} < b_{n+1} < b_n$ strictly for all n and $b_n - a_n \rightarrow 0$, prove the common point ξ satisfies $a_n < \xi < b_n$ strictly for every n .⁷²

SOURCE: ECNU 习题 7.1 #3 (bk. p. 155).

S2.10. Construct a divergent sequence (a_n) with the property that for every $\varepsilon > 0$ there is an N such that $n > N$ implies $|a_{n+1} - a_n| < \varepsilon$.⁷³

SOURCE: USTC Ch. 1 suppl. #4 (bk. p. 38).

S2.11. $\star$ (The completion, built by hand.) Let (X, d) be a metric space. Call two Cauchy sequences in X equivalent if $d(x_n, y_n) \rightarrow 0$, write $\bar{X}$ for the set of equivalence classes, and define $\bar{d}([x], [y]) = \lim d(x_n, y_n)$. Prove: (a) the relation is an equivalence relation; (b) the limit defining $\bar{d}$ exists and is independent of representatives, and $\bar{d}$ is a metric; (c) $\bar{X}$ is complete; (d) $x \mapsto [(x, x, x, \dots)]$ embeds X isometrically; (e) the image is dense in $\bar{X}$.⁷⁴

SOURCE: Tao II, Ex. 1.4.8 (bk. p. 21); Yu Pin 定理 77 (p. 123) adds the universal property.

S2.12. Prove that a metric space is compact if and only if it is complete and totally bounded (S2.9).⁷⁵

SOURCE: Tao II, Ex. 1.5.10 (bk. p. 26).

S2.13. In the space X of absolutely summable sequences with $d(a, b) = \sum_n |a_n - b_n|$, let $e^{(n)}$ be the sequence with 1 in place n and 0 elsewhere. Prove that $\{e^{(n)} : n \in \mathbb{N}\}$ is closed and bounded, that X is complete, and that the set is nevertheless not compact.⁷⁶

SOURCE: Tao II, Ex. 1.5.8 with Ex. 1.1.15 (bk. pp. 26, 10).

S2.14. (Contraction mapping principle.) Let (X, d) be complete and $T : X \rightarrow X$ satisfy $d(Tx, Ty) \leq \gamma d(x, y)$ for some fixed $\gamma \in (0, 1)$. Prove T has exactly one fixed point. Then prove the refinement: if some iterate T^r is a contraction, T still has exactly one fixed point — and check that $x \mapsto e^{-x}$ on $[0, \infty)$ is not a contraction although $f \circ f$ is.⁷⁷

SOURCE: Yu Pin 定理 37 (p. 77) and the following exercises.

69 Rudin's 4.19, but assigned here — and the point of assigning it here is that Chapter 2's machinery is already enough. Each x gives δ_x for $\varepsilon/2$; the balls of radius $\delta_x/2$ cover $[a, b]$; take a finite subcover and let δ be the smallest radius. The halving is what makes two nearby points land in a common ball. Compare S4.19 in the Chapter 4 appendix, which names the δ this construction produces.

70 Monotonicity collapses the whole cluster set to a point — the sequence version of the fact that a monotone function has one-sided limits everywhere. Contrast with $\sin(n\pi/4)$ at S2.2, which has five: order is exactly what forbids returning.

71 The concrete instance of “[a, b] is perfect,” which Rudin asserts in passing at 2.44's discussion but never asks anyone to verify. Both inclusions need a word: every point of $[a, b]$ is approached from within, and no point outside is.

72 S2.3 gives only $a_n \leq \xi \leq b_n$, and several later arguments need the strict version — in particular any argument that must place ξ in the interior of every stage. Cheap to prove, easy to assume without noticing.

73 The pseudo-Cauchy counterexample: consecutive terms getting close does not make a sequence Cauchy. The canonical answer is the partial sums of the harmonic series, where $|a_{n+1} - a_n| = 1/(n+1) \rightarrow 0$ while the sums diverge. Rudin states 3.10 with the full $|a_n - a_m|$ and never asks why the weaker condition will not do — but every student proposes it eventually.

74 The construction of $\mathbb{R}$ from $\mathbb{Q}$, done once in generality — Rudin gives it only as Exercise 3.24 and only sketches it. Part (b) is where the work is: you must prove a limit exists before you can use it as a definition, and that is what the Cauchy hypothesis is for. Chapter 1's appendix builds $\mathbb{R}$ by cuts instead; comparing the two is the best available illustration that “the” real numbers are characterised by a property, not a construction.

75 For the hard direction, build nested subsequences: cover by finitely many balls of radius 1 and keep one containing infinitely many terms; refine with radius 1/2; and so on. The diagonal subsequence is Cauchy, and completeness converts it into a convergent one. The diagonal trick is the same one Rudin uses at 7.23 for Arzelà–Ascoli — worth meeting here first, where it is uncluttered.

76 Sharper than Rudin’s Exercise 2.16, where compactness fails because $\mathbb{Q}$ is incomplete. Here nothing is incomplete: the failure is dimension. Any two of the $e^{(n)}$ are at distance 2, so no subsequence is Cauchy and no finite family of balls of radius 1 covers the set. This is the picture drawn at S2.9.

77 The first theorem in the book that genuinely needs completeness: the iterates form a Cauchy sequence by comparison with a geometric series, and only completeness produces the limit. Rudin proves this at 9.23, five chapters later, for the inverse function theorem; the one-dimensional case is exercise S4.15 in the Chapter 4 appendix.

78 Compactness forcing rigidity. Part (b) is the surprising one — a map that never decreases distances on a compact space cannot increase them either. All three run on the same engine: iterate f , extract a convergent subsequence, and use it to show points that looked separated must be reachable. Nothing in Rudin’s Chapter 2 exercises has this flavour.

79 The universal property of the product, which is what Rudin’s 4.10 (a map into $\mathbb{R}^k$ is continuous iff its components are) is a special case of. The matrix statements are the payoff: openness follows because the determinant is continuous and GL_n is the preimage of $\mathbb{R} \setminus \{0\}$, and continuity of inversion from Cramer’s rule. Chapter 9 will use both without proof.

S2.15. Let (X, d) be a compact metric space and $f : X \rightarrow X$. Prove:
 (a) if $d(f(x), f(y)) = d(x, y)$ for all x, y , then f is a bijection;
 (b) if $d(f(x), f(y)) \geq d(x, y)$ for all x, y , then f is an isometry;
 (c) if $d(f(x), f(y)) \leq d(x, y)$ for all x, y and f is onto, then f is an isometry.⁷⁸

SOURCE: Yu Pin, winter problem S (p. 331).

S2.16. On a product $Y \times Z$ of metric spaces put $d((y_1, z_1), (y_2, z_2)) = \sqrt{d_Y(y_1, y_2)^2 + d_Z(z_1, z_2)^2}$. Prove this is a metric, that the projections are continuous, and that $F : X \rightarrow Y \times Z$ is continuous if and only if both components are. Then prove that the invertible $n \times n$ real matrices form an open subset of all matrices and that $A \mapsto A^{-1}$ is continuous on it.⁷⁹

SOURCE: Yu Pin, homework 4 A8–A12 (p. 108).

S2.17. Prove that $\mathbb{R}^1$ is uncountable using nested intervals rather than a diagonal argument: (a) given a closed interval J of positive length and any x , find a closed $I \subset J$ of positive length with $x \notin I$; (b) given a sequence (x_n) , build nested closed intervals I_n with $x_n \notin I_n$; (c) conclude.⁸⁰

SOURCE: Yu Pin, homework 1 B7 (p. 37).

S2.18. Prove that any family of pairwise disjoint open intervals in $\mathbb{R}^1$ is countable.⁸¹

SOURCE: Yu Pin, homework 1 B11 (p. 37).

S2.19. Prove Cantor’s theorem in general: for any set X there is no surjection $X \rightarrow 2^X$. Then prove the Schröder–Bernstein theorem: if there are injections $A \rightarrow B$ and $B \rightarrow A$, there is a bijection $A \rightarrow B$.⁸²

SOURCE: Tao I, Thm 8.3.1 and Ex. 8.3.2–8.3.3 (bk. pp. 195, 198).

S2.20. ★ Construct a compact topological space containing a sequence with no convergent subsequence.⁸³

SOURCE: Tao II, Ex. 2.5.8 with 2.5.5 (bk. pp. 43, 42).

S2.21. ★ (Furstenberg.) On $\mathbb{Z}$ call a set open if it is empty or a union of arithmetic progressions $U_{a,b} = \{ka + b : k \in \mathbb{Z}\}$ with $a \geq 1$. Prove: (a) this is a topology; (b) no nonempty finite set is open, so no set with finite complement is closed; (c) each $U_{a,b}$ is both open and closed; (d) $\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \text{ prime}} U_{p,0}$; (e) conclude that there are infinitely many primes.⁸⁴

SOURCE: Yu Pin, homework 5 娱乐问题 (p. 130).

80 Rudin proves uncountability twice — by diagonalisation at 2.14 and, far more powerfully, as a corollary of “perfect sets are uncountable” at 2.43. This is a third proof, and the cheapest: it uses only S2.3. Note that it is really the same idea as 2.43, with the nested intervals doing what Rudin’s shrinking neighbourhoods do.

81 One line if you see it: each interval contains a rational, and disjointness makes the assignment injective. The technique — inject an unknown family into $\mathbb{Q}$ — is the standard way to prove countability, and it is exactly the argument behind Rudin’s 4.30 (a monotone function has at most countably many discontinuities) and behind Exercise 2.29.

82 Cantor’s proof is Rudin’s 2.14 with the decimals removed:

$A = \{x : x \notin f(x)\}$ is missed by every $f(x)$ — and note its kinship with Russell’s paradox. Schröder–Bernstein is what makes cardinality an order relation; the slick proof partitions A using $D_0 = B \setminus f(A)$ and $D_{n+1} = f(g(D_n))$.

83 The counterexample proving that S2.10 is a theorem about metric spaces and not about topological ones. Tao’s construction well-orders $\mathbb{R}$, takes the union of all countable initial segments, adjoins a maximum, and gives the result its order topology; every point below the maximum has only countably many predecessors, which is what strands the sequence. Rudin never distinguishes the two compactnesses because he never leaves metric spaces.

84 The best advertisement in this chapter for topology as a tool rather than a vocabulary. If there were finitely many primes, the union in (d) would be a finite union of closed sets, hence closed, so $\{-1, 1\}$ would be open — contradicting (b). Every step uses only Rudin’s 2.23–2.24, on a space with no metric in sight.